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(54) **DECIMATED REFERENCE SIGNALING FOR EQUALIZATION**

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(57) **ABSTRACT**

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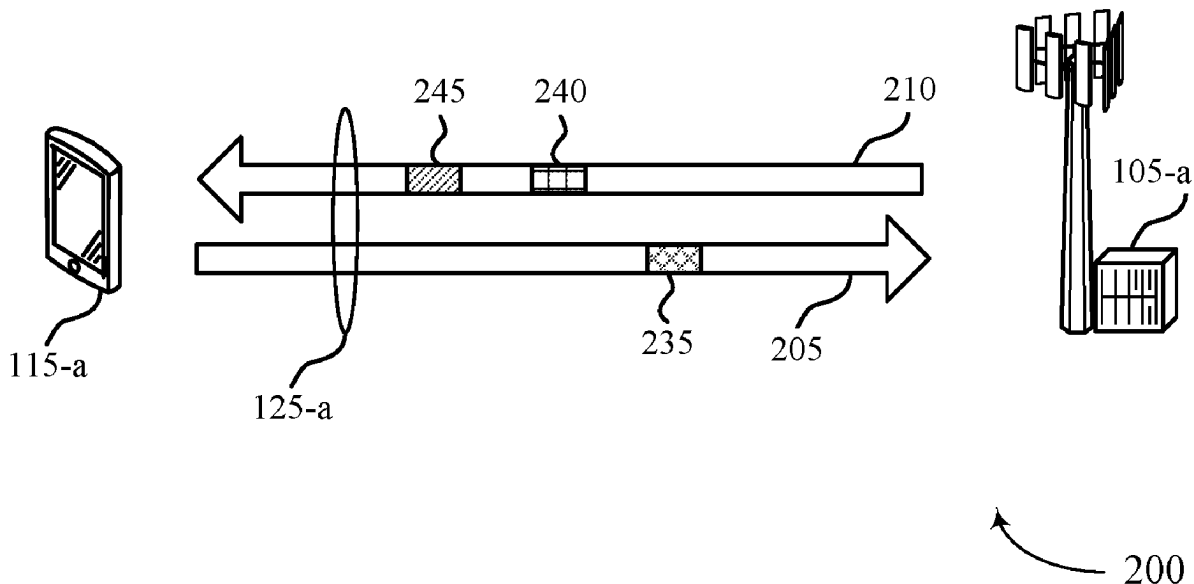
Some wireless communication systems utilize reference signaling to characterize a channel. For example, a network entity may transmit a set of reference signals. A user equipment (UE) may receive the reference signaling and determine a channel estimate based on the reference signaling. The UE may utilize the channel estimate to determine an equalizer for subsequent communications. Some examples of the techniques described herein may reduce complexity or power consumption for equalizer coefficient calculation. In some aspects, reference signaling overhead may be reduced or throughput may be increased. In some examples, decimation may be applied in the calculation of equalizer coefficients. For instance, calculations may be performed for frequency domain or time domain resources, but with a two-dimensional (2D) decimation factor for the two dimensions. In some cases, the 2D decimation factor (e.g., frequency domain and time domain decimation factors) may significantly reduce the quantity of calculations.

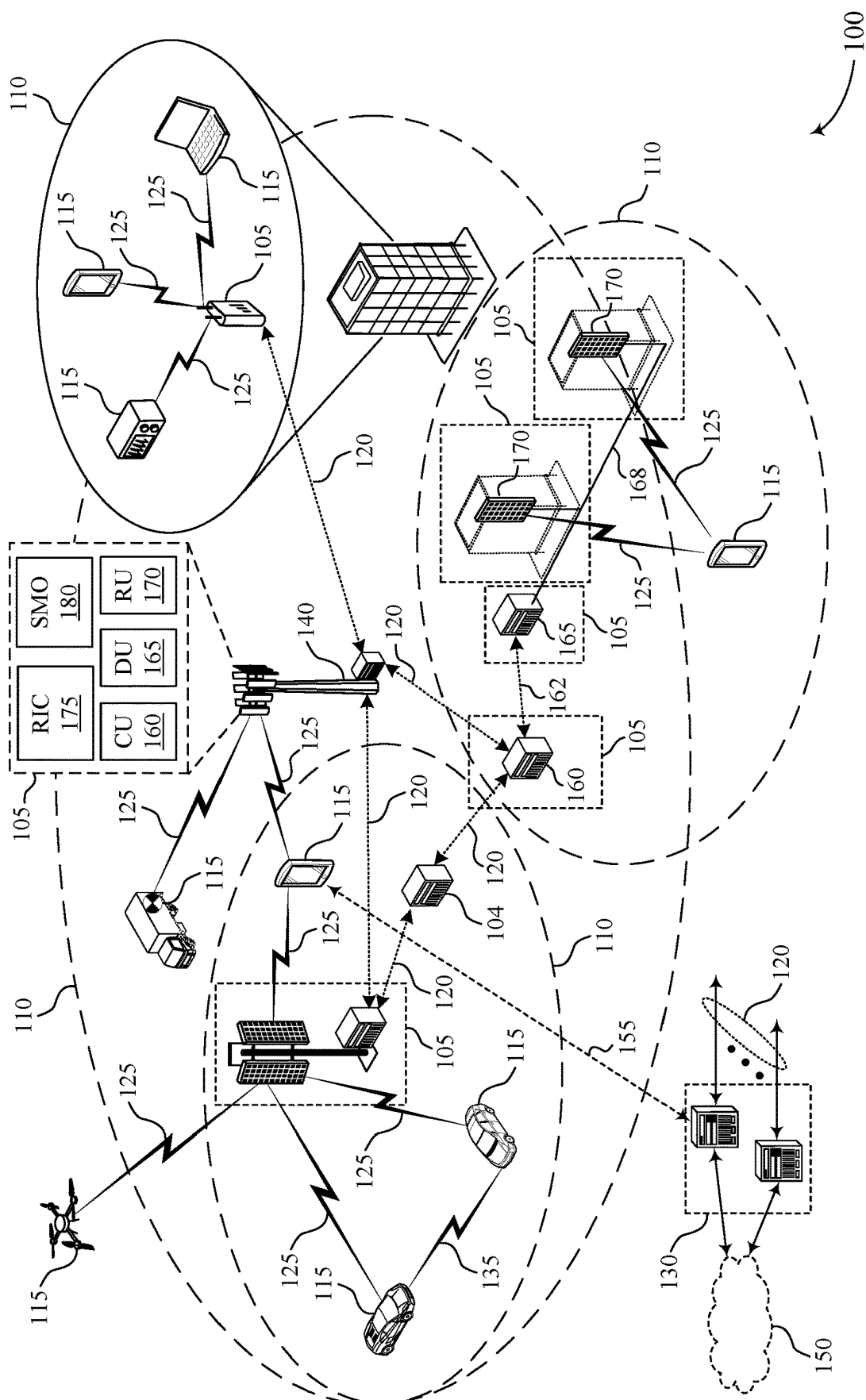
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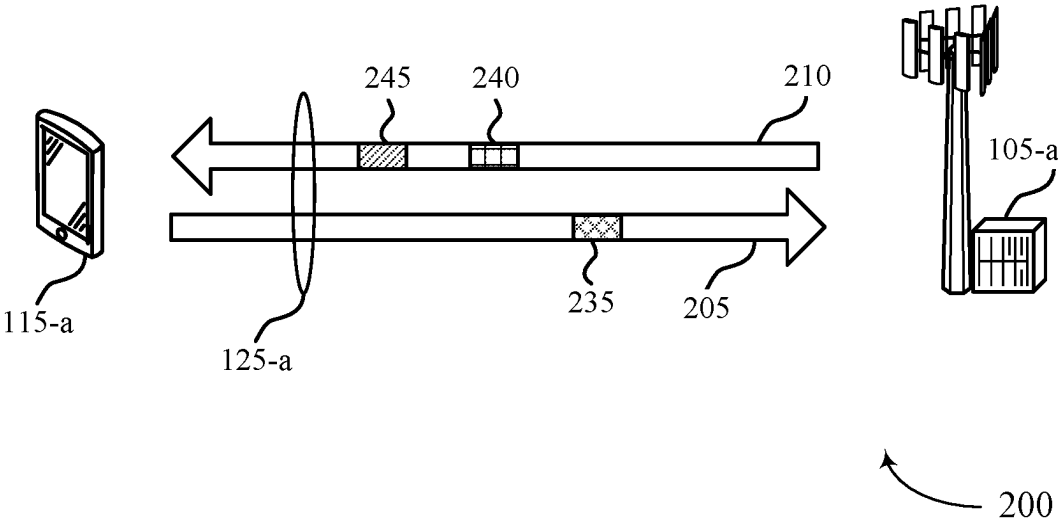


FIG. 2

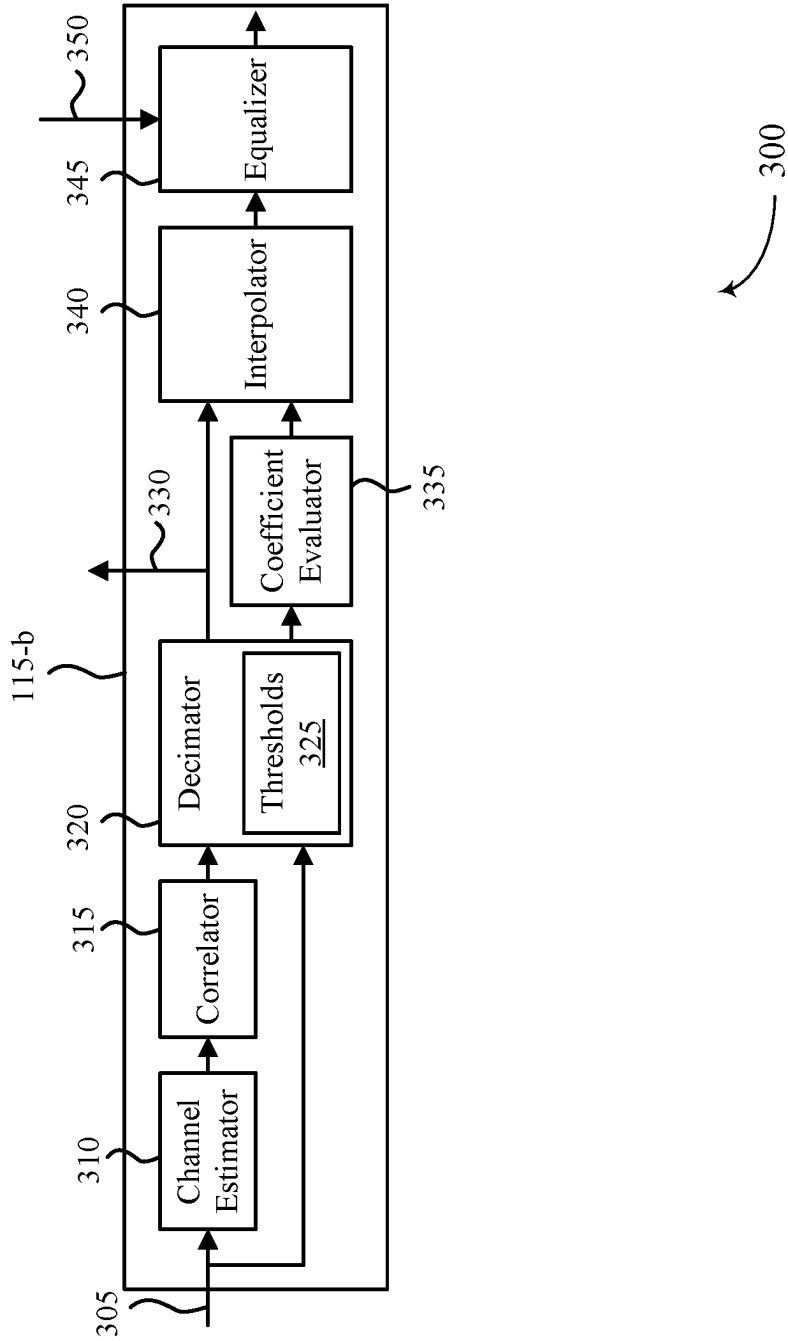


FIG. 3

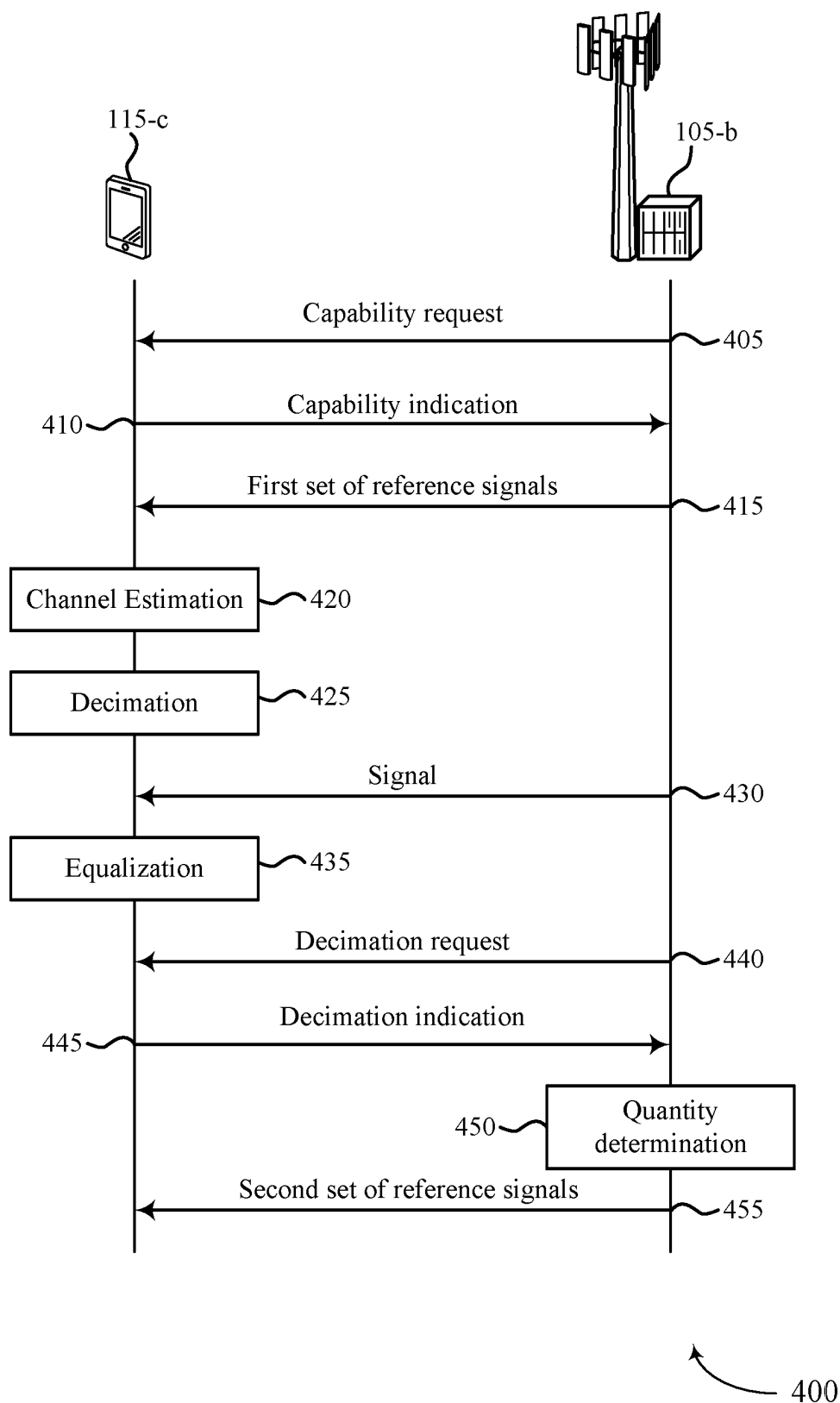
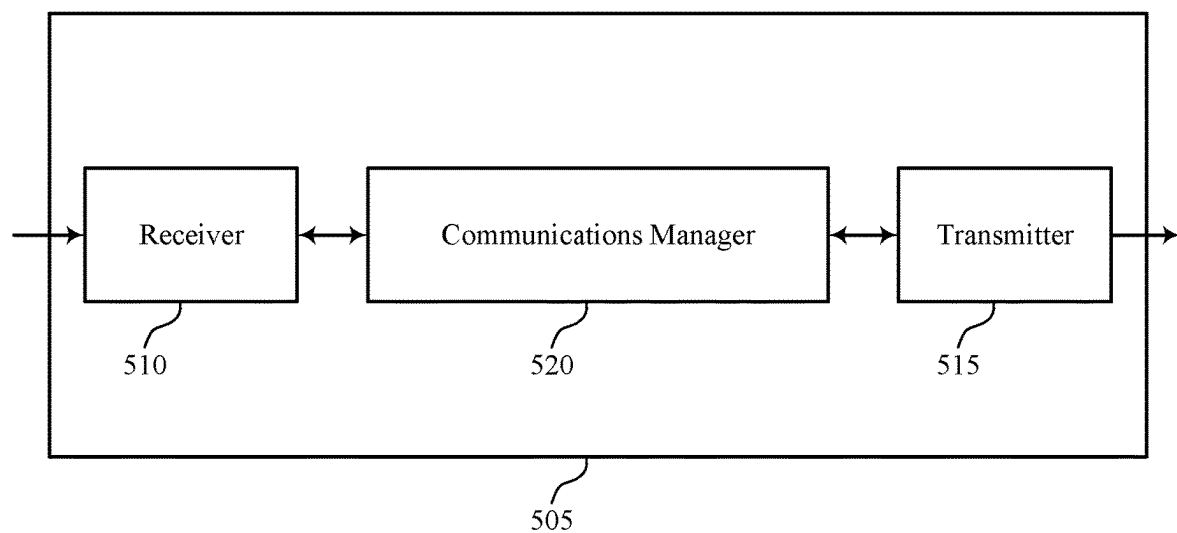


FIG. 4



500

FIG. 5

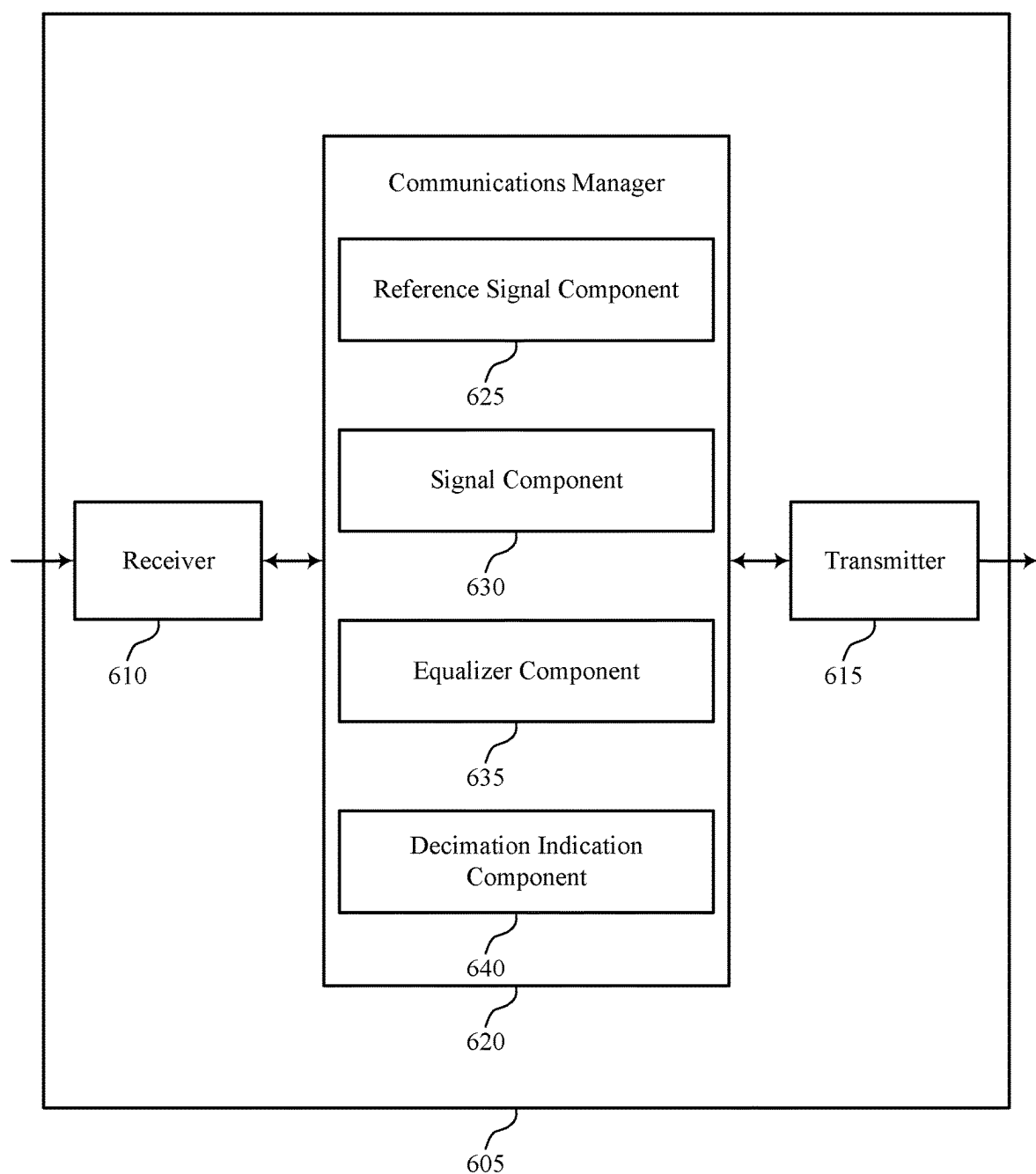


FIG. 6

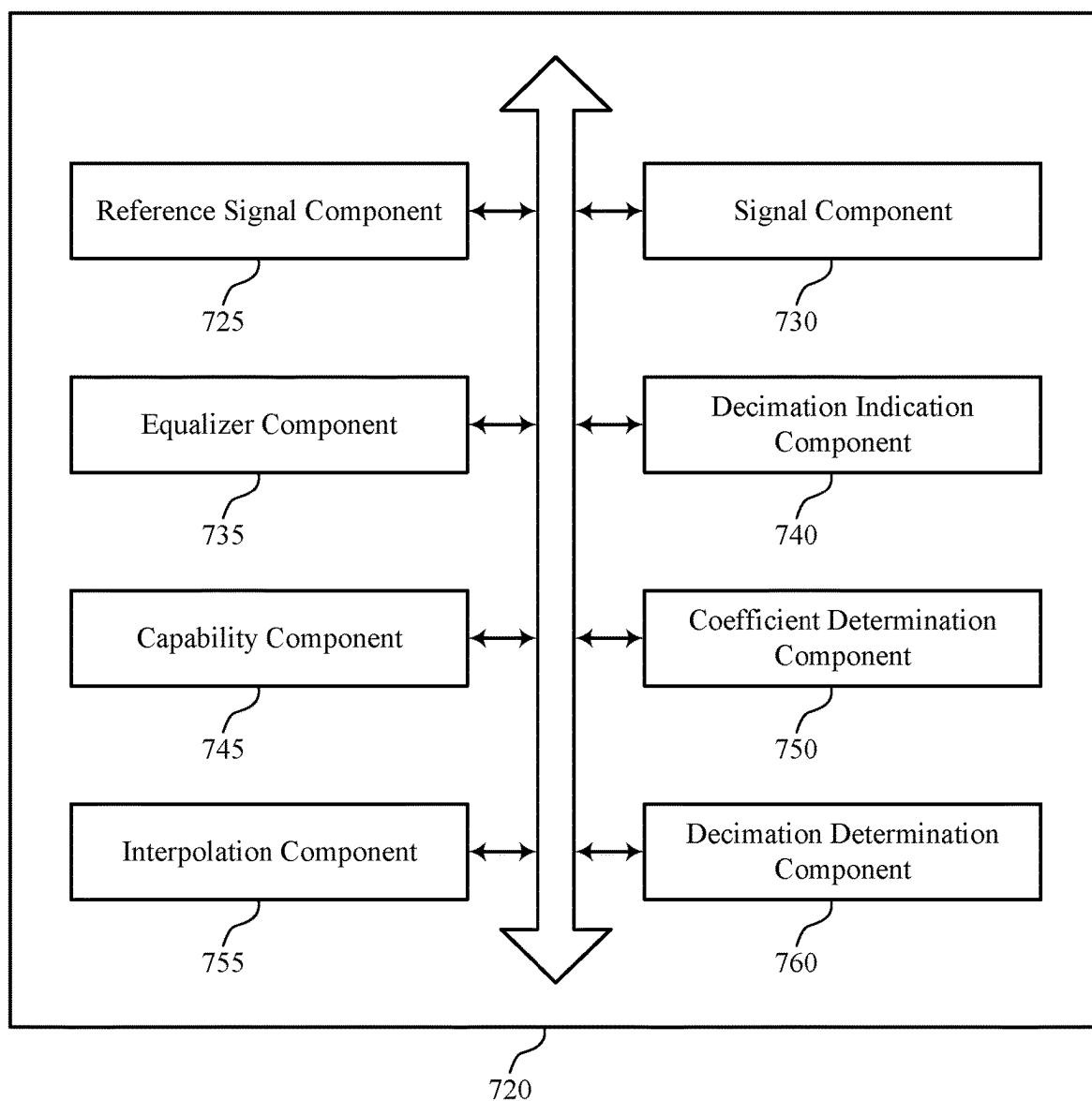


FIG. 7

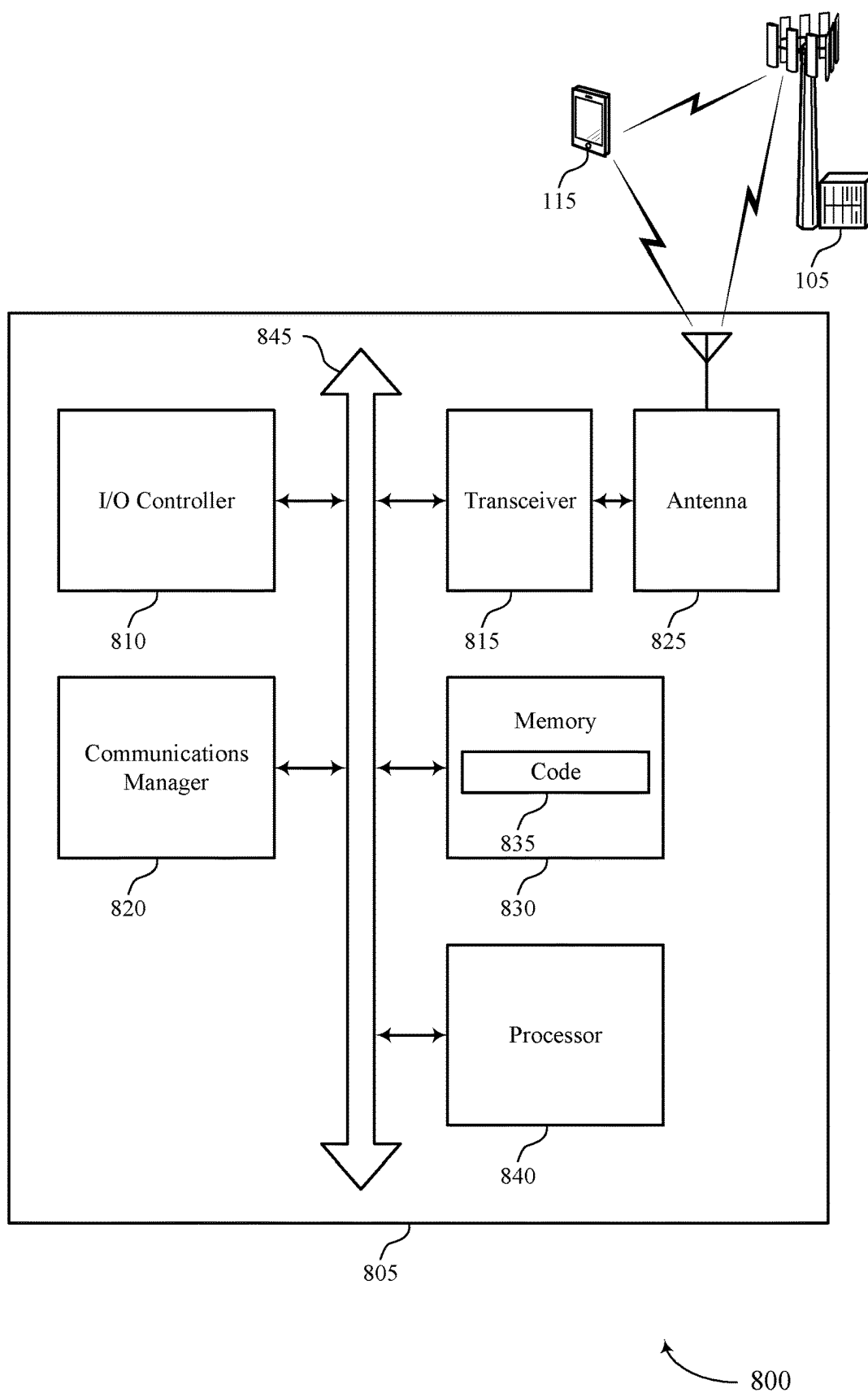


FIG. 8

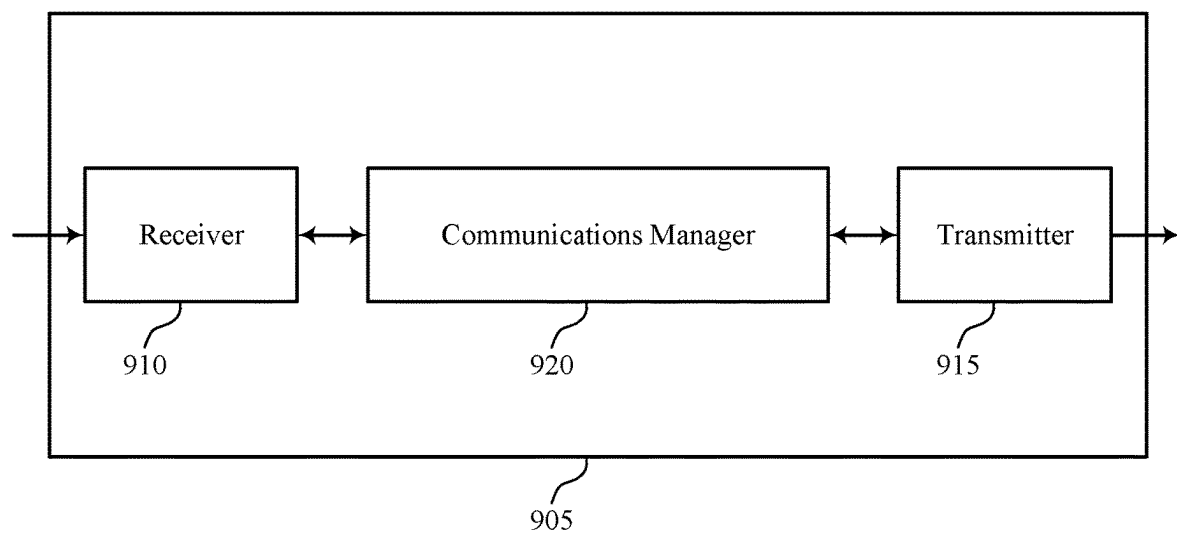


FIG. 9

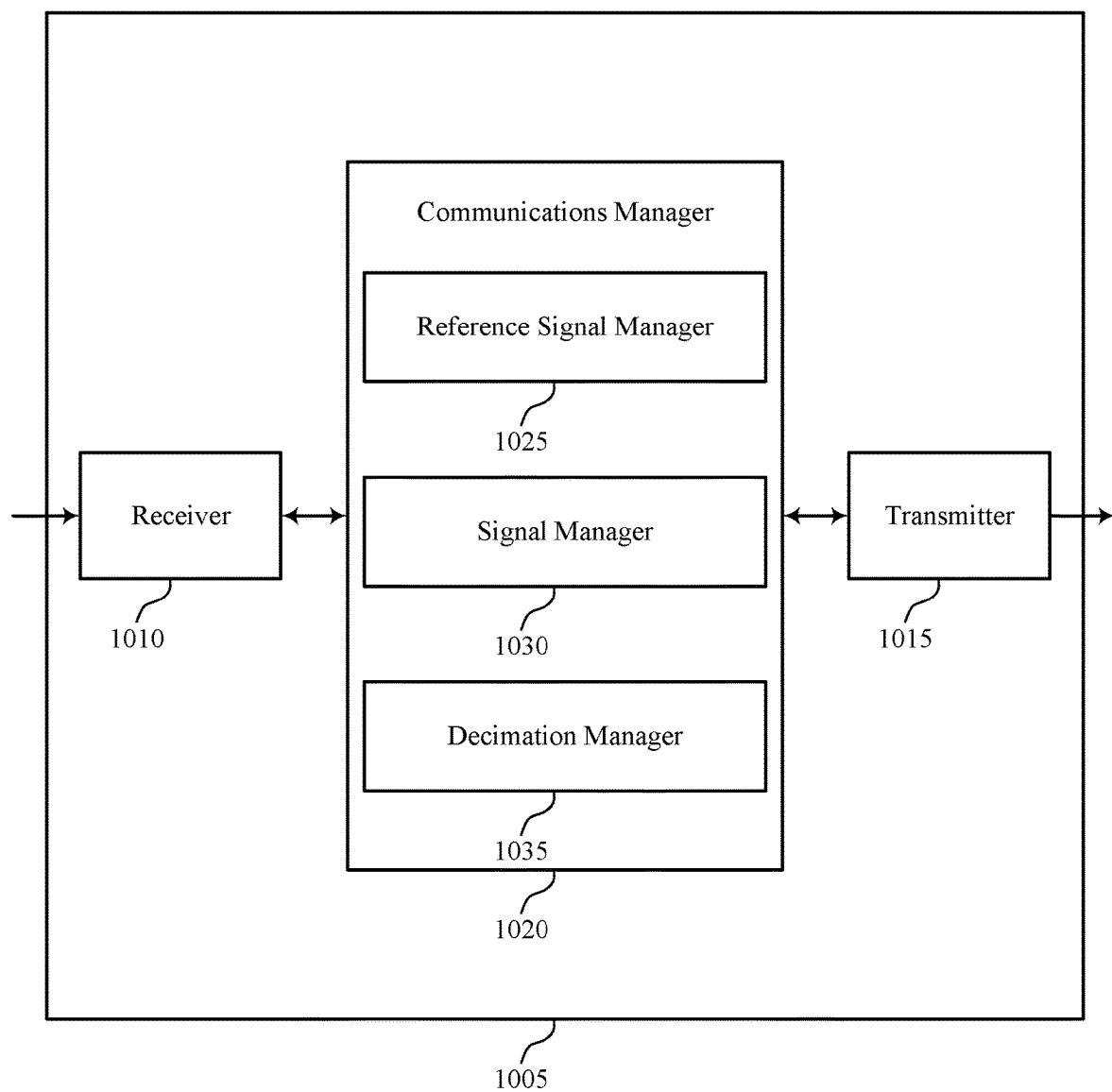
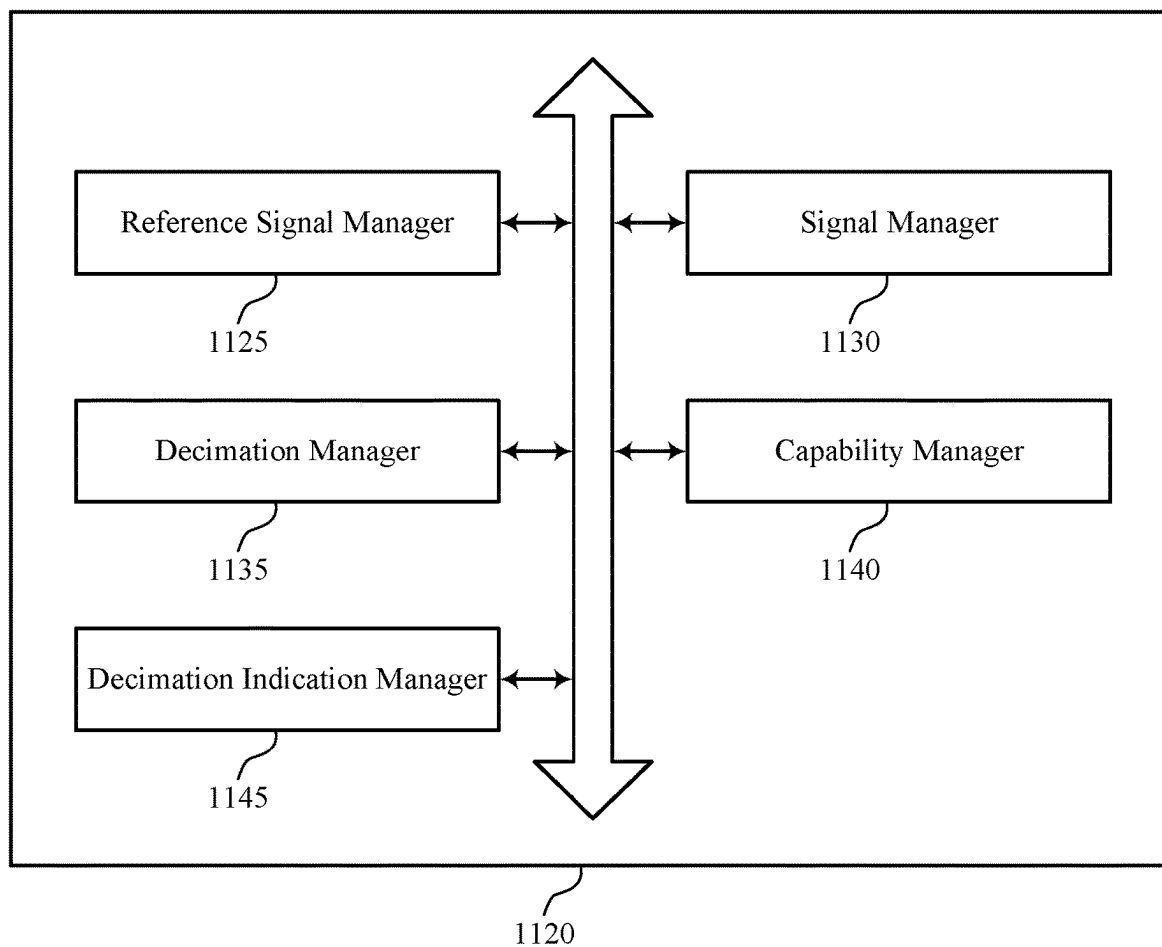
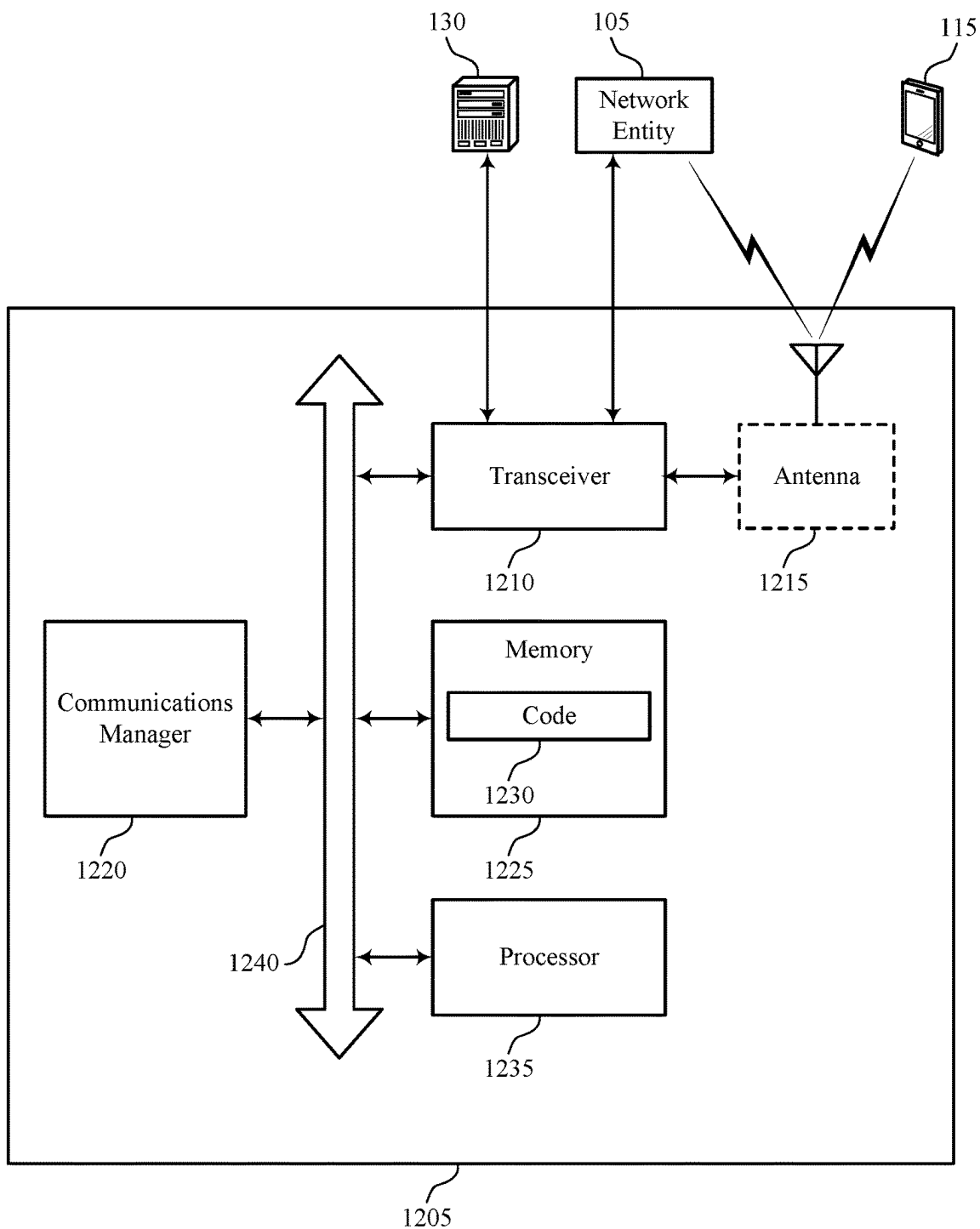


FIG. 10



1100

FIG. 11



1200

FIG. 12

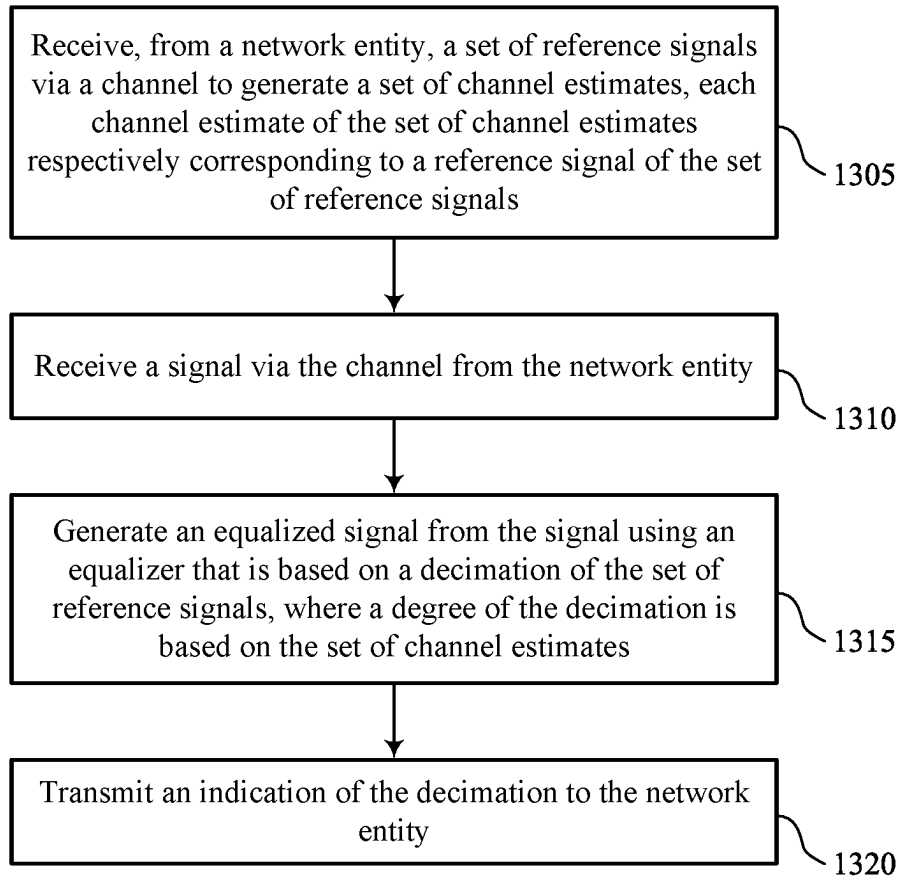
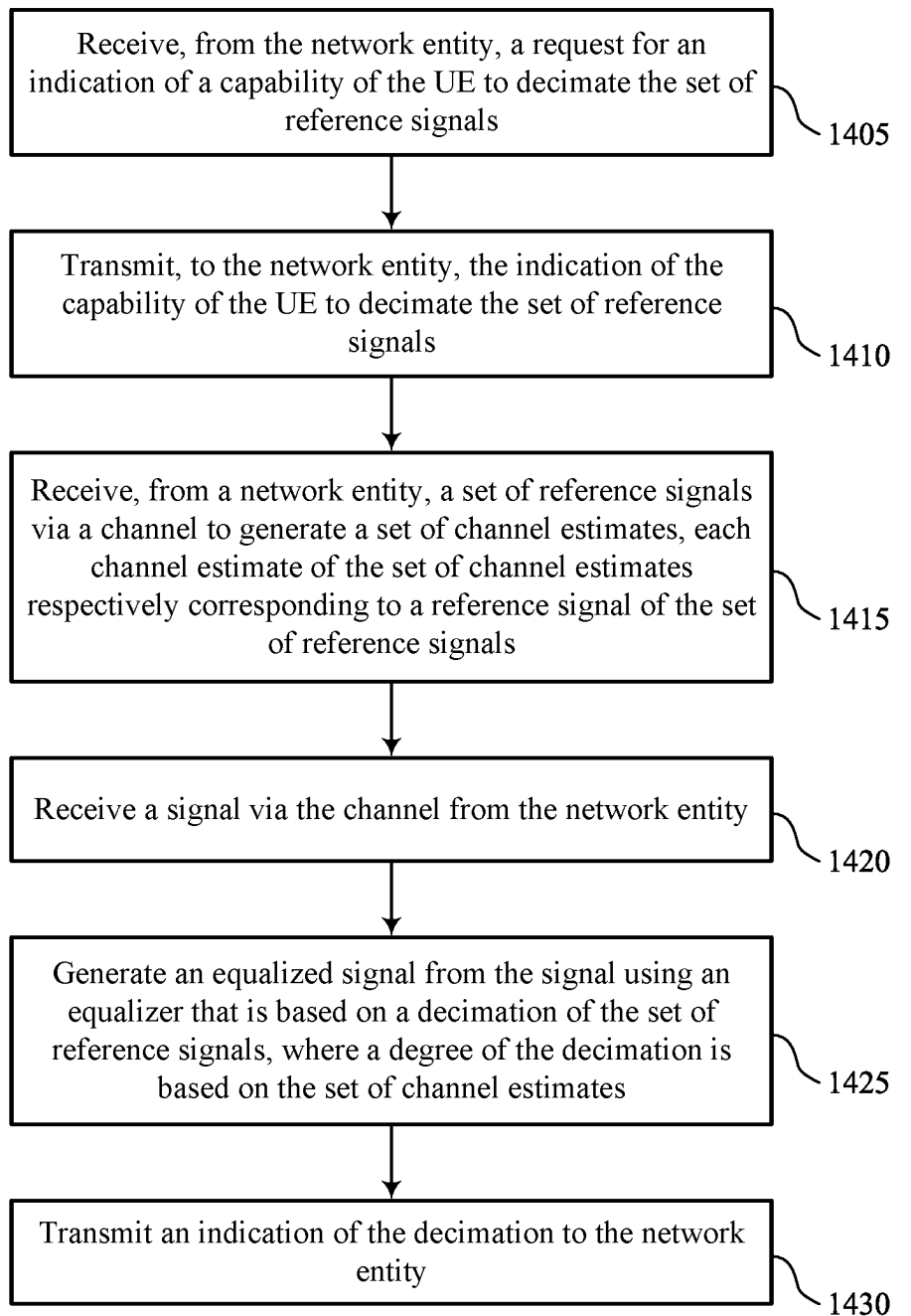
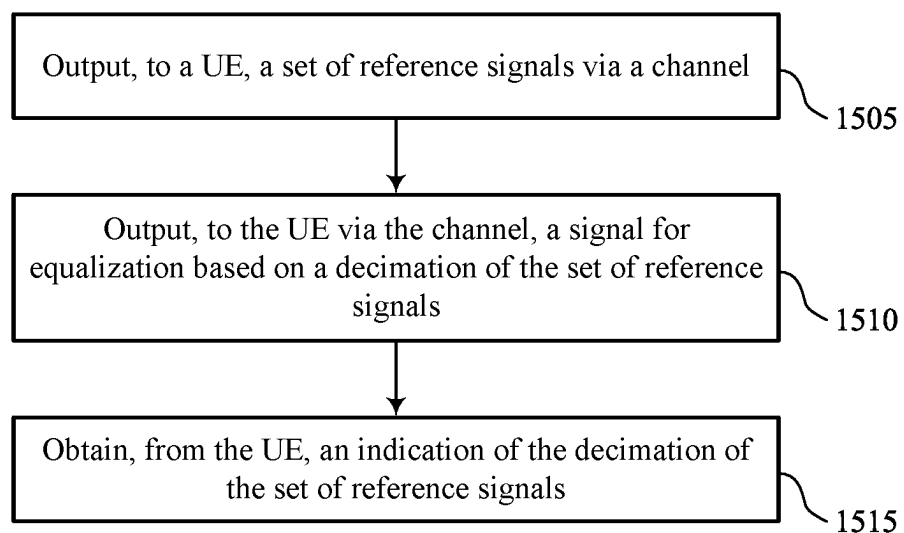


FIG. 13



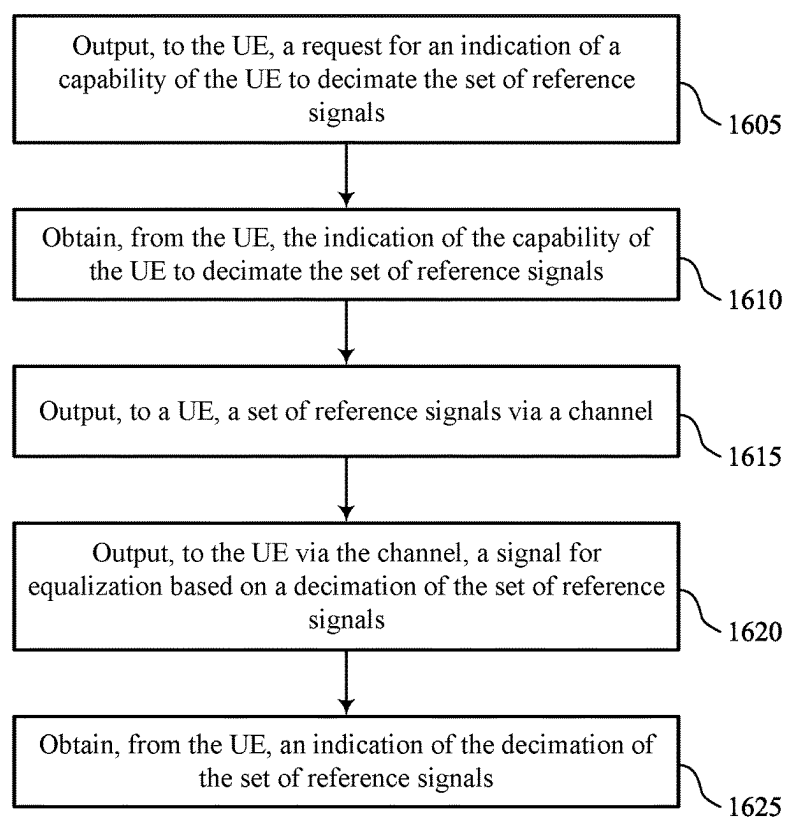
1400

FIG. 14



1500

FIG. 15



1600

FIG. 16

DECIMATED REFERENCE SIGNALING FOR EQUALIZATION

FIELD OF TECHNOLOGY

[0001] The following relates to wireless communications, including decimated reference signaling for equalization.

BACKGROUND

[0002] Wireless communications systems are widely deployed to provide various types of communication content such as voice, video, packet data, messaging, broadcast, and so on. These systems may be capable of supporting communication with multiple users by sharing the available system resources (e.g., time, frequency, and power). Examples of such multiple-access systems include fourth generation (4G) systems such as Long Term Evolution (LTE) systems, LTE-Advanced (LTE-A) systems, or LTE-A Pro systems, and fifth generation (5G) systems which may be referred to as New Radio (NR) systems. These systems may employ technologies such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal FDMA (OFDMA), or discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-S-OFDM). A wireless multiple-access communications system may include one or more base stations, each supporting wireless communication for communication devices, which may be known as user equipment (UE).

SUMMARY

[0003] Some wireless communication systems utilize reference signaling to characterize a channel. For example, a network entity may transmit a set of reference signals. A user equipment (UE) may receive the reference signaling and determine a channel estimate based on the reference signaling. The UE may utilize the channel estimate to determine an equalizer for subsequent communications. Some examples of the techniques described herein may reduce complexity or power consumption for equalizer coefficient calculation. In some aspects, reference signaling (e.g., demodulation reference signal (DMRS)) overhead may be reduced or throughput may be increased. In some examples, decimation may be applied in the calculation of equalizer coefficients. Accordingly, direct coefficient calculation may not be applied per resource (e.g., frequency domain or time domain resource). For example, calculations may be performed for frequency domain or time domain resources, but with a two-dimensional (2D) decimation factor for the two dimensions. In some cases, the 2D decimation factor (e.g., frequency domain and time domain decimation factors) may significantly reduce the quantity of calculations. Accordingly, the power consumption may be significantly reduced. In some approaches, for the frequency domain or time domain resources that were decimated, the equalizer coefficients may be calculated with reduced complexity utilizing a linear interpolation between frequency domain or time domain resources.

[0004] A method by a UE is described. The method may include receiving, from a network entity, a set of reference signals via a channel to generate a set of channel estimates, each channel estimate of the set of channel estimates respectively corresponding to a reference signal of the set of reference signals, receiving a signal via the channel from the

network entity, generating an equalized signal from the signal using an equalizer that is based on a decimation of the set of reference signals, where a degree of the decimation is based on the set of channel estimates, and transmitting an indication of the decimation to the network entity.

[0005] A UE is described. The UE may include one or more memories storing processor executable code, and one or more processors coupled with the one or more memories. The one or more processors may individually or collectively be operable to execute the code to cause the UE to receive, from a network entity, a set of reference signals via a channel to generate a set of channel estimates, each channel estimate of the set of channel estimates respectively corresponding to a reference signal of the set of reference signals, receive a signal via the channel from the network entity, generate an equalized signal from the signal using an equalizer that is based on a decimation of the set of reference signals, where a degree of the decimation is based on the set of channel estimates, and transmit an indication of the decimation to the network entity.

[0006] Another UE is described. The UE may include means for receiving, from a network entity, a set of reference signals via a channel to generate a set of channel estimates, each channel estimate of the set of channel estimates respectively corresponding to a reference signal of the set of reference signals, means for receiving a signal via the channel from the network entity, means for generating an equalized signal from the signal using an equalizer that is based on a decimation of the set of reference signals, where a degree of the decimation is based on the set of channel estimates, and means for transmitting an indication of the decimation to the network entity.

[0007] A non-transitory computer-readable medium storing code is described. The code may include instructions executable by one or more processors to receive, from a network entity, a set of reference signals via a channel to generate a set of channel estimates, each channel estimate of the set of channel estimates respectively corresponding to a reference signal of the set of reference signals, receive a signal via the channel from the network entity, generate an equalized signal from the signal using an equalizer that is based on a decimation of the set of reference signals, where a degree of the decimation is based on the set of channel estimates, and transmit an indication of the decimation to the network entity.

[0008] Some examples of the method, UEs, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from the network entity, a request for an indication of a capability of the UE to decimate the set of reference signals and transmitting, to the network entity, the indication of the capability of the UE to decimate the set of reference signals.

[0009] Some examples of the method, UEs, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from the network entity, a request for the indication of the decimation, where the indication of the decimation may be transmitted in response to the request.

[0010] Some examples of the method, UEs, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from the network entity, a second quantity of a second set of reference signals, where the second quantity

may be different from a first quantity of the set of reference signals, where the second quantity may be based on the indication of the decimation.

[0011] Some examples of the method, UEs, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining a first set of coefficients corresponding to a first reference signal received via a first resource of the channel, determining a second set of coefficients corresponding to a second reference signal received via a second resource of the channel, and interpolating the first set of coefficients and the second set of coefficients for a third resource of the channel between the first resource and the second resource to generate a third set of coefficients of the equalizer, where a third reference signal corresponding to the third resource may be decimated in accordance with the decimation.

[0012] Some examples of the method, UEs, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining the degree of decimation based on the set of reference signals.

[0013] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, determining the degree of decimation may include operations, features, means, or instructions for generating a measure of correlation between at least two channel estimates of the set of channel estimates.

[0014] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the measure of correlation includes a first measure in a time domain, a second measure in a frequency domain, or a combination thereof.

[0015] Some examples of the method, UEs, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for selecting the degree of decimation based on at least one threshold applied to the measure of correlation.

[0016] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the at least one threshold includes a first set of thresholds for a time domain and a second set of thresholds for a frequency domain.

[0017] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the at least one threshold includes multiple sets of thresholds corresponding to respective modulation and coding schemes (MCSs).

[0018] A method by a network entity is described. The method may include outputting, to a UE, a set of reference signals via a channel, outputting, to the UE via the channel, a signal for equalization based on a decimation of the set of reference signals, and obtaining, from the UE, an indication of the decimation of the set of reference signals.

[0019] A network entity is described. The network entity may include one or more memories storing processor executable code, and one or more processors coupled with the one or more memories. The one or more processors may individually or collectively be operable to execute the code to cause the network entity to output, to a UE, a set of reference signals via a channel, output, to the UE via the channel, a signal for equalization based on a decimation of the set of reference signals, and obtain, from the UE, an indication of the decimation of the set of reference signals.

[0020] Another network entity is described. The network entity may include means for outputting, to a UE, a set of reference signals via a channel, means for outputting, to the UE via the channel, a signal for equalization based on a decimation of the set of reference signals, and means for obtaining, from the UE, an indication of the decimation of the set of reference signals.

[0021] A non-transitory computer-readable medium storing code is described. The code may include instructions executable by one or more processors to output, to a UE, a set of reference signals via a channel, output, to the UE via the channel, a signal for equalization based on a decimation of the set of reference signals, and obtain, from the UE, an indication of the decimation of the set of reference signals.

[0022] Some examples of the method, network entities, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for outputting, to the UE, a request for an indication of a capability of the UE to decimate the set of reference signals and obtaining, from the UE, the indication of the capability of the UE to decimate the set of reference signals.

[0023] Some examples of the method, network entities, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for outputting, to the UE, a request for the indication of the decimation, where the indication of the decimation may be obtained in response to the request.

[0024] Some examples of the method, network entities, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining, based on the indication of the decimation, a quantity for a second set of reference signals.

[0025] Some examples of the method, network entities, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for outputting, to the UE, a second quantity of the second set of reference signals, where the second quantity may be different from a first quantity of the set of reference signals, where the second quantity may be based on the quantity.

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] FIG. 1 shows an example of a wireless communications system that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure.

[0027] FIG. 2 shows an example of a wireless communications system that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure.

[0028] FIG. 3 shows an example of a block diagram that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure.

[0029] FIG. 4 shows an example of a process flow that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure.

[0030] FIGS. 5 and 6 show block diagrams of devices that support decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure.

[0031] FIG. 7 shows a block diagram of a communications manager that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure.

[0032] FIG. 8 shows a diagram of a system including a device that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure.

[0033] FIGS. 9 and 10 show block diagrams of devices that support decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure.

[0034] FIG. 11 shows a block diagram of a communications manager that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure.

[0035] FIG. 12 shows a diagram of a system including a device that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure.

[0036] FIGS. 13 through 16 show flowcharts illustrating methods that support decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure.

DETAILED DESCRIPTION

[0037] Some wireless communication systems utilize reference signaling to characterize a channel. For example, a network entity may transmit a set of reference signals. A user equipment (UE) may receive the reference signaling and determine a channel estimate based on the reference signaling. The UE may utilize the channel estimate to determine an equalizer for subsequent communications.

[0038] One challenge in wireless communications is to reduce the power consumption of the UE. Equalizer coefficient calculation may consume a relatively large amount of processing resources at the UE receiver. In some approaches, the equalizer (e.g., linear minimum mean-squared error (LMMSE) equalizer) coefficients are calculated per each frequency domain or time domain resource in a slot. Accordingly, a relatively large quantity of calculations may be carried out for coefficient calculation. Moreover, a calculation for a single frequency domain or time domain resource may be relatively costly regarding the quantity of multiplications because the calculation may involve matrix inversion and matrix multiplication. Further, some UEs may support a relatively higher quantity of layers (e.g., spatial layers or multiple-input multiple-output (MIMO) layers). For example, instead of four layers, eight layers may be utilized. Accordingly, UEs may have a greater quantity of receive antennas (e.g., eight antennas instead of four antennas) to accommodate more layers. Because the quantity of receive antennas may be utilized to determine the channel dimensions (e.g., H dimensions), the equalizer coefficient calculation complexity may increase with increasing channel dimensions. Even without an increasing the quantity of layers, power consumption for equalizer coefficients calculation is significant.

[0039] In some examples, decimation may be applied in the calculation of equalizer coefficients to reduce calculation complexity or power consumption. For instance, direct coefficient calculation may not be applied per resource (e.g., frequency domain or time domain resource). Calculations may be performed for frequency domain or time domain

resources, but with one or more decimation factors. For example, a two-dimensional (2D) decimation factor may be utilized for frequency and time dimensions. In some cases, a 2D decimation factor (e.g., frequency domain and time domain decimation factors) may significantly reduce the quantity of calculations. Accordingly, calculation complexity or power consumption may be significantly reduced. In some approaches, for the frequency domain or time domain resources that were decimated, the equalizer coefficients may be calculated with reduced complexity utilizing a linear interpolation between frequency domain or time domain resources.

[0040] Some examples of the techniques described herein may be utilized to reduce the complexity or power consumption of equalizer coefficient calculation. Some approaches may preserve performance (e.g., avoid or reduce performance loss) while reducing complexity or power consumption. In some aspects, reference signaling (e.g., demodulation reference signal (DMRS)) overhead may be reduced or throughput may be increased.

[0041] Aspects of the disclosure are initially described in the context of wireless communications systems. Aspects of the disclosure are additionally described in the context of a block diagram and process flow. Aspects of the disclosure are further illustrated by and described with reference to apparatus diagrams, system diagrams, and flowcharts that relate to decimated reference signaling for equalization.

[0042] FIG. 1 shows an example of a wireless communications system 100 that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure. The wireless communications system 100 may include one or more devices, such as one or more network devices (e.g., network entities 105), one or more UEs 115, and a core network 130. In some examples, the wireless communications system 100 may be a Long Term Evolution (LTE) network, an LTE-Advanced (LTE-A) network, an LTE-A Pro network, a New Radio (NR) network, or a network operating in accordance with other systems and radio technologies, including future systems and radio technologies not explicitly mentioned herein.

[0043] The network entities 105 may be dispersed throughout a geographic area to form the wireless communications system 100 and may include devices in different forms or having different capabilities. In various examples, a network entity 105 may be referred to as a network element, a mobility element, a radio access network (RAN) node, or network equipment, among other nomenclature. In some examples, network entities 105 and UEs 115 may wirelessly communicate via communication link(s) 125 (e.g., a radio frequency (RF) access link). For example, a network entity 105 may support a coverage area 110 (e.g., a geographic coverage area) over which the UEs 115 and the network entity 105 may establish the communication link(s) 125. The coverage area 110 may be an example of a geographic area over which a network entity 105 and a UE 115 may support the communication of signals according to one or more radio access technologies (RATs).

[0044] The UEs 115 may be dispersed throughout a coverage area 110 of the wireless communications system 100, and each UE 115 may be stationary, or mobile, or both at different times. The UEs 115 may be devices in different forms or having different capabilities. Some example UEs 115 are illustrated in FIG. 1. The UEs 115 described herein may be capable of supporting communications with various

types of devices in the wireless communications system 100 (e.g., other wireless communication devices, including UEs 115 or network entities 105), as shown in FIG. 1.

[0045] As described herein, a node of the wireless communications system 100, which may be referred to as a network node, or a wireless node, may be a network entity 105 (e.g., any network entity described herein), a UE 115 (e.g., any UE described herein), a network controller, an apparatus, a device, a computing system, one or more components, or another suitable processing entity configured to perform any of the techniques described herein. For example, a node may be a UE 115. As another example, a node may be a network entity 105. As another example, a first node may be configured to communicate with a second node or a third node. In one aspect of this example, the first node may be a UE 115, the second node may be a network entity 105, and the third node may be a UE 115. In another aspect of this example, the first node may be a UE 115, the second node may be a network entity 105, and the third node may be a network entity 105. In yet other aspects of this example, the first, second, and third nodes may be different relative to these examples. Similarly, reference to a UE 115, network entity 105, apparatus, device, computing system, or the like may include disclosure of the UE 115, network entity 105, apparatus, device, computing system, or the like being a node. For example, disclosure that a UE 115 is configured to receive information from a network entity 105 also discloses that a first node is configured to receive information from a second node.

[0046] In some examples, network entities 105 may communicate with a core network 130, or with one another, or both. For example, network entities 105 may communicate with the core network 130 via backhaul communication link(s) 120 (e.g., in accordance with an S1, N2, N3, or other interface protocol). In some examples, network entities 105 may communicate with one another via backhaul communication link(s) 120 (e.g., in accordance with an X2, Xn, or other interface protocol) either directly (e.g., directly between network entities 105) or indirectly (e.g., via the core network 130). In some examples, network entities 105 may communicate with one another via a midhaul communication link 162 (e.g., in accordance with a midhaul interface protocol) or a fronthaul communication link 168 (e.g., in accordance with a fronthaul interface protocol), or any combination thereof. The backhaul communication link(s) 120, midhaul communication links 162, or fronthaul communication links 168 may be or include one or more wired links (e.g., an electrical link, an optical fiber link) or one or more wireless links (e.g., a radio link, a wireless optical link), among other examples or various combinations thereof. A UE 115 may communicate with the core network 130 via a communication link 155.

[0047] One or more of the network entities 105 or network equipment described herein may include or may be referred to as a base station 140 (e.g., a base transceiver station, a radio base station, an NR base station, an access point, a radio transceiver, a NodeB, an eNodeB (eNB), a next-generation NodeB or giga-NodeB (either of which may be referred to as a gNB), a 5G NB, a next-generation eNB (ng-eNB), a Home NodeB, a Home eNodeB, or other suitable terminology). In some examples, a network entity 105 (e.g., a base station 140) may be implemented in an aggregated (e.g., monolithic, standalone) base station architecture, which may be configured to utilize a protocol stack

that is physically or logically integrated within one network entity (e.g., a network entity 105 or a single RAN node, such as a base station 140).

[0048] In some examples, a network entity 105 may be implemented in a disaggregated architecture (e.g., a disaggregated base station architecture, a disaggregated RAN architecture), which may be configured to utilize a protocol stack that is physically or logically distributed among multiple network entities (e.g., network entities 105), such as an integrated access and backhaul (IAB) network, an open RAN (O-RAN) (e.g., a network configuration sponsored by the O-RAN Alliance), or a virtualized RAN (vRAN) (e.g., a cloud RAN (C-RAN)). For example, a network entity 105 may include one or more of a central unit (CU), such as a CU 160, a distributed unit (DU), such as a DU 165, a radio unit (RU), such as an RU 170, a RAN Intelligent Controller (RIC), such as an RIC 175 (e.g., a Near-Real Time RIC (Near-RT RIC), a Non-Real Time RIC (Non-RT RIC)), a Service Management and Orchestration (SMO) system, such as an SMO system 180, or any combination thereof. An RU 170 may also be referred to as a radio head, a smart radio head, a remote radio head (RRH), a remote radio unit (RRU), or a transmission reception point (TRP). One or more components of the network entities 105 in a disaggregated RAN architecture may be co-located, or one or more components of the network entities 105 may be located in distributed locations (e.g., separate physical locations). In some examples, one or more of the network entities 105 of a disaggregated RAN architecture may be implemented as virtual units (e.g., a virtual CU (VCU), a virtual DU (VDU), a virtual RU (VRU)).

[0049] The split of functionality between a CU 160, a DU 165, and an RU 170 is flexible and may support different functionalities depending on which functions (e.g., network layer functions, protocol layer functions, baseband functions, RF functions, or any combinations thereof) are performed at a CU 160, a DU 165, or an RU 170. For example, a functional split of a protocol stack may be employed between a CU 160 and a DU 165 such that the CU 160 may support one or more layers of the protocol stack and the DU 165 may support one or more different layers of the protocol stack. In some examples, the CU 160 may host upper protocol layer (e.g., layer 3 (L3), layer 2 (L2)) functionality and signaling (e.g., Radio Resource Control (RRC), service data adaptation protocol (SDAP), Packet Data Convergence Protocol (PDCP)). The CU 160 (e.g., one or more CUs) may be connected to a DU 165 (e.g., one or more DUs) or an RU 170 (e.g., one or more RUs), or some combination thereof, and the DUs 165, RUs 170, or both may host lower protocol layers, such as layer 1 (L1) (e.g., physical (PHY) layer) or L2 (e.g., radio link control (RLC) layer, medium access control (MAC) layer) functionality and signaling, and may each be at least partially controlled by the CU 160. Additionally, or alternatively, a functional split of the protocol stack may be employed between a DU 165 and an RU 170 such that the DU 165 may support one or more layers of the protocol stack and the RU 170 may support one or more different layers of the protocol stack. The DU 165 may support one or multiple different cells (e.g., via one or multiple different RUs, such as an RU 170). In some cases, a functional split between a CU 160 and a DU 165 or between a DU 165 and an RU 170 may be within a protocol layer (e.g., some functions for a protocol layer may be performed by one of a CU 160, a DU 165, or an RU 170,

while other functions of the protocol layer are performed by a different one of the CU 160, the DU 165, or the RU 170). A CU 160 may be functionally split further into CU control plane (CU-CP) and CU user plane (CU-UP) functions. A CU 160 may be connected to a DU 165 via a midhaul communication link 162 (e.g., F1, F1-c, F1-u), and a DU 165 may be connected to an RU 170 via a fronthaul communication link 168 (e.g., open fronthaul (FH) interface). In some examples, a midhaul communication link 162 or a fronthaul communication link 168 may be implemented in accordance with an interface (e.g., a channel) between layers of a protocol stack supported by respective network entities (e.g., one or more of the network entities 105) that are in communication via such communication links.

[0050] In some wireless communications systems (e.g., the wireless communications system 100), infrastructure and spectral resources for radio access may support wireless backhaul link capabilities to supplement wired backhaul connections, providing an IAB network architecture (e.g., to a core network 130). In some cases, in an IAB network, one or more of the network entities 105 (e.g., network entities 105 or IAB node(s) 104) may be partially controlled by each other. The IAB node(s) 104 may be referred to as a donor entity or an IAB donor. A DU 165 or an RU 170 may be partially controlled by a CU 160 associated with a network entity 105 or base station 140 (such as a donor network entity or a donor base station). The one or more donor entities (e.g., IAB donors) may be in communication with one or more additional devices (e.g., IAB node(s) 104) via supported access and backhaul links (e.g., backhaul communication link(s) 120). IAB node(s) 104 may include an IAB mobile termination (IAB-MT) controlled (e.g., scheduled) by one or more DUs (e.g., DUs 165) of a coupled IAB donor. An IAB-MT may be equipped with an independent set of antennas for relay of communications with UEs 115 or may share the same antennas (e.g., of an RU 170) of IAB node(s) 104 used for access via the DU 165 of the IAB node(s) 104 (e.g., referred to as virtual IAB-MT (vIAB-MT)). In some examples, the IAB node(s) 104 may include one or more DUs (e.g., DUs 165) that support communication links with additional entities (e.g., IAB node(s) 104, UEs 115) within the relay chain or configuration of the access network (e.g., downstream). In such cases, one or more components of the disaggregated RAN architecture (e.g., the IAB node(s) 104 or components of the IAB node(s) 104) may be configured to operate according to the techniques described herein.

[0051] For instance, an access network (AN) or RAN may include communications between access nodes (e.g., an IAB donor), IAB node(s) 104, and one or more UEs 115. The IAB donor may facilitate connection between the core network 130 and the AN (e.g., via a wired or wireless connection to the core network 130). That is, an IAB donor may refer to a RAN node with a wired or wireless connection to the core network 130. The IAB donor may include one or more of a CU 160, a DU 165, and an RU 170, in which case the CU 160 may communicate with the core network 130 via an interface (e.g., a backhaul link). The IAB donor and IAB node(s) 104 may communicate via an F1 interface according to a protocol that defines signaling messages (e.g., an F1 AP protocol). Additionally, or alternatively, the CU 160 may communicate with the core network 130 via an interface, which may be an example of a portion of a backhaul link, and may communicate with

other CUs (e.g., including a CU 160 associated with an alternative IAB donor) via an Xn-C interface, which may be an example of another portion of a backhaul link.

[0052] IAB node(s) 104 may refer to RAN nodes that provide IAB functionality (e.g., access for UEs 115, wireless self-backhauling capabilities). A DU 165 may act as a distributed scheduling node towards child nodes associated with the IAB node(s) 104, and the IAB-MT may act as a scheduled node towards parent nodes associated with IAB node(s) 104. That is, an IAB donor may be referred to as a parent node in communication with one or more child nodes (e.g., an IAB donor may relay transmissions for UEs through other IAB node(s) 104). Additionally, or alternatively, IAB node(s) 104 may also be referred to as parent nodes or child nodes to other IAB node(s) 104, depending on the relay chain or configuration of the AN. The IAB-MT entity of IAB node(s) 104 may provide a Uu interface for a child IAB node (e.g., the IAB node(s) 104) to receive signaling from a parent IAB node (e.g., the IAB node(s) 104), and a DU interface (e.g., a DU 165) may provide a Uu interface for a parent IAB node to signal to a child IAB node or UE 115.

[0053] For example, IAB node(s) 104 may be referred to as parent nodes that support communications for child IAB nodes, or may be referred to as child IAB nodes associated with IAB donors, or both. An IAB donor may include a CU 160 with a wired or wireless connection (e.g., backhaul communication link(s) 120) to the core network 130 and may act as a parent node to IAB node(s) 104. For example, the DU 165 of an IAB donor may relay transmissions to UEs 115 through IAB node(s) 104, or may directly signal transmissions to a UE 115, or both. The CU 160 of the IAB donor may signal communication link establishment via an F1 interface to IAB node(s) 104, and the IAB node(s) 104 may schedule transmissions (e.g., transmissions to the UEs 115 relayed from the IAB donor) through one or more DUs (e.g., DUs 165). That is, data may be relayed to and from IAB node(s) 104 via signaling via an NR Uu interface to MT of IAB node(s) 104 (e.g., other IAB node(s)). Communications with IAB node(s) 104 may be scheduled by a DU 165 of the IAB donor or of IAB node(s) 104.

[0054] In the case of the techniques described herein applied in the context of a disaggregated RAN architecture, one or more components of the disaggregated RAN architecture may be configured to support test as described herein. For example, some operations described as being performed by a UE 115 or a network entity 105 (e.g., a base station 140) may additionally, or alternatively, be performed by one or more components of the disaggregated RAN architecture (e.g., components such as an IAB node, a DU 165, a CU 160, an RU 170, an RIC 175, an SMO system 180).

[0055] A UE 115 may include or may be referred to as a mobile device, a wireless device, a remote device, a handheld device, or a subscriber device, or some other suitable terminology, where the “device” may also be referred to as a unit, a station, a terminal, or a client, among other examples. A UE 115 may also include or may be referred to as a personal electronic device such as a cellular phone, a personal digital assistant (PDA), a tablet computer, a laptop computer, or a personal computer. In some examples, a UE 115 may include or be referred to as a wireless local loop (WLL) station, an Internet of Things (IoT) device, an Internet of Everything (IoE) device, or a machine type communications (MTC) device, among other examples,

which may be implemented in various objects such as appliances, vehicles, or meters, among other examples.

[0056] The UEs **115** described herein may be able to communicate with various types of devices, such as UEs **115** that may sometimes operate as relays, as well as the network entities **105** and the network equipment including macro eNBs or gNBs, small cell eNBs or gNBs, or relay base stations, among other examples, as shown in FIG. 1.

[0057] The UEs **115** and the network entities **105** may wirelessly communicate with one another via the communication link(s) **125** (e.g., one or more access links) using resources associated with one or more carriers. The term “carrier” may refer to a set of RF spectrum resources having a defined PHY layer structure for supporting the communication link(s) **125**. For example, a carrier used for the communication link(s) **125** may include a portion of an RF spectrum band (e.g., a bandwidth part (BWP)) that is operated according to one or more PHY layer channels for a given RAT (e.g., LTE, LTE-A, LTE-A Pro, NR). Each PHY layer channel may carry acquisition signaling (e.g., synchronization signals, system information), control signaling that coordinates operation for the carrier, user data, or other signaling. The wireless communications system **100** may support communication with a UE **115** using carrier aggregation or multi-carrier operation. A UE **115** may be configured with multiple downlink component carriers and one or more uplink component carriers according to a carrier aggregation configuration. Carrier aggregation may be used with both frequency division duplexing (FDD) and time division duplexing (TDD) component carriers. Communication between a network entity **105** and other devices may refer to communication between the devices and any portion (e.g., entity, sub-entity) of a network entity **105**. For example, the terms “transmitting,” “receiving,” or “communicating,” when referring to a network entity **105**, may refer to any portion of a network entity **105** (e.g., a base station **140**, a CU **160**, a DU **165**, a RU **170**) of a RAN communicating with another device (e.g., directly or via one or more other network entities, such as one or more of the network entities **105**).

[0058] In some examples, such as in a carrier aggregation configuration, a carrier may have acquisition signaling or control signaling that coordinates operations for other carriers. A carrier may be associated with a frequency channel (e.g., an evolved universal mobile telecommunication system terrestrial radio access (E-UTRA) absolute RF channel number (EARFCN)) and may be identified according to a channel raster for discovery by the UEs **115**. A carrier may be operated in a standalone mode, in which case initial acquisition and connection may be conducted by the UEs **115** via the carrier, or the carrier may be operated in a non-standalone mode, in which case a connection is anchored using a different carrier (e.g., of the same or a different RAT).

[0059] The communication link(s) **125** of the wireless communications system **100** may include downlink transmissions (e.g., forward link transmissions) from a network entity **105** to a UE **115**, uplink transmissions (e.g., return link transmissions) from a UE **115** to a network entity **105**, or both, among other configurations of transmissions. Carriers may carry downlink or uplink communications (e.g., in an FDD mode) or may be configured to carry downlink and uplink communications (e.g., in a TDD mode).

[0060] A carrier may be associated with a particular bandwidth of the RF spectrum and, in some examples, the carrier bandwidth may be referred to as a “system bandwidth” of the carrier or the wireless communications system **100**. For example, the carrier bandwidth may be one of a set of bandwidths for carriers of a particular RAT (e.g., 1.4, 3, 5, 10, 15, 20, 40, or 80 megahertz (MHz)). Devices of the wireless communications system **100** (e.g., the network entities **105**, the UEs **115**, or both) may have hardware configurations that support communications using a particular carrier bandwidth or may be configurable to support communications using one of a set of carrier bandwidths. In some examples, the wireless communications system **100** may include network entities **105** or UEs **115** that support concurrent communications using carriers associated with multiple carrier bandwidths. In some examples, each served UE **115** may be configured for operating using portions (e.g., a sub-band, a BWP) or all of a carrier bandwidth.

[0061] Signal waveforms transmitted via a carrier may be made up of multiple subcarriers (e.g., using multi-carrier modulation (MCM) techniques such as orthogonal frequency division multiplexing (OFDM) or discrete Fourier transform spread OFDM (DFT-S-OFDM)). In a system employing MCM techniques, a resource element may refer to resources of one symbol period (e.g., a duration of one modulation symbol) and one subcarrier, in which case the symbol period and subcarrier spacing may be inversely related. The quantity of bits carried by each resource element may depend on the modulation scheme (e.g., the order of the modulation scheme, the coding rate of the modulation scheme, or both), such that a relatively higher quantity of resource elements (e.g., in a transmission duration) and a relatively higher order of a modulation scheme may correspond to a relatively higher rate of communication. A wireless communications resource may refer to a combination of an RF spectrum resource, a time resource, and a spatial resource (e.g., a spatial layer, a beam), and the use of multiple spatial resources may increase the data rate or data integrity for communications with a UE **115**.

[0062] One or more numerologies for a carrier may be supported, and a numerology may include a subcarrier spacing (Δf) and a cyclic prefix. A carrier may be divided into one or more BWPs having the same or different numerologies. In some examples, a UE **115** may be configured with multiple BWPs. In some examples, a single BWP for a carrier may be active at a given time and communications for the UE **115** may be restricted to one or more active BWPs.

[0063] The time intervals for the network entities **105** or the UEs **115** may be expressed in multiples of a basic time unit which may, for example, refer to a sampling period of $T_s = 1/(\Delta f_{max} \cdot N_f)$ seconds, for which Δf_{max} may represent a supported subcarrier spacing, and N_f may represent a supported discrete Fourier transform (DFT) size. Time intervals of a communications resource may be organized according to radio frames each having a specified duration (e.g., 10 milliseconds (ms)). Each radio frame may be identified by a system frame number (SFN) (e.g., ranging from 0 to 1023).

[0064] Each frame may include multiple consecutively-numbered subframes or slots, and each subframe or slot may have the same duration. In some examples, a frame may be divided (e.g., in the time domain) into subframes, and each subframe may be further divided into a quantity of slots. Alternatively, each frame may include a variable quantity of

slots, and the quantity of slots may depend on subcarrier spacing. Each slot may include a quantity of symbol periods (e.g., depending on the length of the cyclic prefix prepended to each symbol period). In some wireless communications systems, such as the wireless communications system **100**, a slot may further be divided into multiple mini-slots associated with one or more symbols. Excluding the cyclic prefix, each symbol period may be associated with one or more (e.g., N_p) sampling periods. The duration of a symbol period may depend on the subcarrier spacing or frequency band of operation.

[0065] A subframe, a slot, a mini-slot, or a symbol may be the smallest scheduling unit (e.g., in the time domain) of the wireless communications system **100** and may be referred to as a transmission time interval (TTI). In some examples, the TTI duration (e.g., a quantity of symbol periods in a TTI) may be variable. Additionally, or alternatively, the smallest scheduling unit of the wireless communications system **100** may be dynamically selected (e.g., in bursts of shortened TTIs (sTTIs)).

[0066] Physical channels may be multiplexed for communication using a carrier according to various techniques. A physical control channel and a physical data channel may be multiplexed for signaling via a downlink carrier, for example, using one or more of time division multiplexing (TDM) techniques, frequency division multiplexing (FDM) techniques, or hybrid TDM-FDM techniques. A control region (e.g., a control resource set (CORESET)) for a physical control channel may be defined by a set of symbol periods and may extend across the system bandwidth or a subset of the system bandwidth of the carrier. One or more control regions (e.g., CORESETs) may be configured for a set of the UEs **115**. For example, one or more of the UEs **115** may monitor or search control regions for control information according to one or more search space sets, and each search space set may include one or multiple control channel candidates in one or more aggregation levels arranged in a cascaded manner. An aggregation level for a control channel candidate may refer to an amount of control channel resources (e.g., control channel elements (CCEs)) associated with encoded information for a control information format having a given payload size. Search space sets may include common search space sets configured for sending control information to UEs **115** (e.g., one or more UEs) or may include UE-specific search space sets for sending control information to a UE **115** (e.g., a specific UE).

[0067] A network entity **105** may provide communication coverage via one or more cells, for example a macro cell, a small cell, a hot spot, or other types of cells, or any combination thereof. The term “cell” may refer to a logical communication entity used for communication with a network entity **105** (e.g., using a carrier) and may be associated with an identifier for distinguishing neighboring cells (e.g., a physical cell identifier (PCID), a virtual cell identifier (VCID)). In some examples, a cell also may refer to a coverage area **110** or a portion of a coverage area **110** (e.g., a sector) over which the logical communication entity operates. Such cells may range from smaller areas (e.g., a structure, a subset of structure) to larger areas depending on various factors such as the capabilities of the network entity **105**. For example, a cell may be or include a building, a subset of a building, or exterior spaces between or overlapping with coverage areas **110**, among other examples.

[0068] A macro cell generally covers a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by the UEs **115** with service subscriptions with the network provider supporting the macro cell. A small cell may be associated with a network entity **105** operating with lower power (e.g., a base station **140** operating with lower power) relative to a macro cell, and a small cell may operate using the same or different (e.g., licensed, unlicensed) frequency bands as macro cells. Small cells may provide unrestricted access to the UEs **115** with service subscriptions with the network provider or may provide restricted access to the UEs **115** having an association with the small cell (e.g., the UEs **115** in a closed subscriber group (CSG), the UEs **115** associated with users in a home or office). A network entity **105** may support one or more cells and may also support communications via the one or more cells using one or multiple component carriers.

[0069] In some examples, a carrier may support multiple cells, and different cells may be configured according to different protocol types (e.g., MTC, narrowband IoT (NB-IoT), enhanced mobile broadband (eMBB)) that may provide access for different types of devices.

[0070] In some examples, a network entity **105** (e.g., a base station **140**, an RU **170**) may be movable and therefore provide communication coverage for a moving coverage area, such as the coverage area **110**. In some examples, coverage areas **110** (e.g., different coverage areas) associated with different technologies may overlap, but the coverage areas **110** (e.g., different coverage areas) may be supported by the same network entity (e.g., a network entity **105**). In some other examples, overlapping coverage areas, such as a coverage area **110**, associated with different technologies may be supported by different network entities (e.g., the network entities **105**). The wireless communications system **100** may include, for example, a heterogeneous network in which different types of the network entities **105** support communications for coverage areas **110** (e.g., different coverage areas) using the same or different RATs.

[0071] The wireless communications system **100** may support synchronous or asynchronous operation. For synchronous operation, network entities **105** (e.g., base stations **140**) may have similar frame timings, and transmissions from different network entities (e.g., different ones of the network entities **105**) may be approximately aligned in time. For asynchronous operation, network entities **105** may have different frame timings, and transmissions from different network entities (e.g., different ones of network entities **105**) may, in some examples, not be aligned in time. The techniques described herein may be used for either synchronous or asynchronous operations.

[0072] Some UEs **115**, such as MTC or IoT devices, may be relatively low cost or low complexity devices and may provide for automated communication between machines (e.g., via Machine-to-Machine (M2M) communication). M2M communication or MTC may refer to data communication technologies that allow devices to communicate with one another or a network entity **105** (e.g., a base station **140**) without human intervention. In some examples, M2M communication or MTC may include communications from devices that integrate sensors or meters to measure or capture information and relay such information to a central server or application program that uses the information or presents the information to humans interacting with the application program. Some UEs **115** may be designed to

collect information or enable automated behavior of machines or other devices. Examples of applications for MTC devices include smart metering, inventory monitoring, water level monitoring, equipment monitoring, healthcare monitoring, wildlife monitoring, weather and geological event monitoring, fleet management and tracking, remote security sensing, physical access control, and transaction-based business charging.

[0073] Some UEs 115 may be configured to employ operating modes that reduce power consumption, such as half-duplex communications (e.g., a mode that supports one-way communication via transmission or reception, but not transmission and reception concurrently). In some examples, half-duplex communications may be performed at a reduced peak rate. Other power conservation techniques for the UEs 115 may include entering a power saving deep sleep mode when not engaging in active communications, operating using a limited bandwidth (e.g., according to narrowband communications), or a combination of these techniques. For example, some UEs 115 may be configured for operation using a narrowband protocol type that is associated with a defined portion or range (e.g., set of subcarriers or resource blocks (RBs)) within a carrier, within a guard-band of a carrier, or outside of a carrier.

[0074] The wireless communications system 100 may be configured to support ultra-reliable communications or low-latency communications, or various combinations thereof. For example, the wireless communications system 100 may be configured to support ultra-reliable low-latency communications (URLLC). The UEs 115 may be designed to support ultra-reliable, low-latency, or critical functions. Ultra-reliable communications may include private communication or group communication and may be supported by one or more services such as push-to-talk, video, or data. Support for ultra-reliable, low-latency functions may include prioritization of services, and such services may be used for public safety or general commercial applications. The terms ultra-reliable, low-latency, and ultra-reliable low-latency may be used interchangeably herein.

[0075] In some examples, a UE 115 may be configured to support communicating directly with other UEs (e.g., one or more of the UEs 115) via a device-to-device (D2D) communication link, such as a D2D communication link 135 (e.g., in accordance with a peer-to-peer (P2P), D2D, or sidelink protocol). In some examples, one or more UEs 115 of a group that are performing D2D communications may be within the coverage area 110 of a network entity 105 (e.g., a base station 140, an RU 170), which may support aspects of such D2D communications being configured by (e.g., scheduled by) the network entity 105. In some examples, one or more UEs 115 of such a group may be outside the coverage area 110 of a network entity 105 or may be otherwise unable to or not configured to receive transmissions from a network entity 105. In some examples, groups of the UEs 115 communicating via D2D communications may support a one-to-many (1:M) system in which each UE 115 transmits to one or more of the UEs 115 in the group. In some examples, a network entity 105 may facilitate the scheduling of resources for D2D communications. In some other examples, D2D communications may be carried out between the UEs 115 without an involvement of a network entity 105.

[0076] In some systems, a D2D communication link 135 may be an example of a communication channel, such as a

sidelink communication channel, between vehicles (e.g., UEs 115). In some examples, vehicles may communicate using vehicle-to-everything (V2X) communications, vehicle-to-vehicle (V2V) communications, or some combination of these. A vehicle may signal information related to traffic conditions, signal scheduling, weather, safety, emergencies, or any other information relevant to a V2X system. In some examples, vehicles in a V2X system may communicate with roadside infrastructure, such as roadside units, or with the network via one or more network nodes (e.g., network entities 105, base stations 140, RUs 170) using vehicle-to-network (V2N) communications, or with both.

[0077] The core network 130 may provide user authentication, access authorization, tracking, Internet Protocol (IP) connectivity, and other access, routing, or mobility functions. The core network 130 may be an evolved packet core (EPC) or 5G core (5GC), which may include at least one control plane entity that manages access and mobility (e.g., a mobility management entity (MME), an access and mobility management function (AMF)) and at least one user plane entity that routes packets or interconnects to external networks (e.g., a serving gateway (S-GW), a Packet Data Network (PDN) gateway (P-GW), or a user plane function (UPF)). The control plane entity may manage non-access stratum (NAS) functions such as mobility, authentication, and bearer management for the UEs 115 served by the network entities 105 (e.g., base stations 140) associated with the core network 130. User IP packets may be transferred through the user plane entity, which may provide IP address allocation as well as other functions. The user plane entity may be connected to IP services 150 for one or more network operators. The IP services 150 may include access to the Internet, Intranet(s), an IP Multimedia Subsystem (IMS), or a Packet-Switched Streaming Service.

[0078] The wireless communications system 100 may operate using one or more frequency bands, which may be in the range of 300 megahertz (MHz) to 300 gigahertz (GHz). Generally, the region from 300 MHz to 3 GHz is known as the ultra-high frequency (UHF) region or decimeter band because the wavelengths range from approximately one decimeter to one meter in length. UHF waves may be blocked or redirected by buildings and environmental features, which may be referred to as clusters, but the waves may penetrate structures sufficiently for a macro cell to provide service to the UEs 115 located indoors. Communications using UHF waves may be associated with smaller antennas and shorter ranges (e.g., less than one hundred kilometers) compared to communications using the smaller frequencies and longer waves of the high frequency (HF) or very high frequency (VHF) portion of the spectrum below 300 MHz.

[0079] The wireless communications system 100 may also operate using a super high frequency (SHF) region, which may be in the range of 3 GHz to 30 GHz, also known as the centimeter band, or using an extremely high frequency (EHF) region of the spectrum (e.g., from 30 GHz to 300 GHz), also known as the millimeter band. In some examples, the wireless communications system 100 may support millimeter wave (mmW) communications between the UEs 115 and the network entities 105 (e.g., base stations 140, RUs 170), and EHF antennas of the respective devices may be smaller and more closely spaced than UHF antennas. In some examples, such techniques may facilitate using antenna arrays within a device. The propagation of EHF

transmissions, however, may be subject to even greater attenuation and shorter range than SHF or UHF transmissions. The techniques disclosed herein may be employed across transmissions that use one or more different frequency regions, and designated use of bands across these frequency regions may differ by country or regulating body.

[0080] The wireless communications system **100** may utilize both licensed and unlicensed RF spectrum bands. For example, the wireless communications system **100** may employ License Assisted Access (LAA), LTE-Unlicensed (LTE-U) RAT, or NR technology using an unlicensed band such as the 5 GHz industrial, scientific, and medical (ISM) band. While operating using unlicensed RF spectrum bands, devices such as the network entities **105** and the UEs **115** may employ carrier sensing for collision detection and avoidance. In some examples, operations using unlicensed bands may be based on a carrier aggregation configuration in conjunction with component carriers operating using a licensed band (e.g., LAA). Operations using unlicensed spectrum may include downlink transmissions, uplink transmissions, P2P transmissions, or D2D transmissions, among other examples.

[0081] A network entity **105** (e.g., a base station **140**, an RU **170**) or a UE **115** may be equipped with multiple antennas, which may be used to employ techniques such as transmit diversity, receive diversity, MIMO communications, or beamforming. The antennas of a network entity **105** or a UE **115** may be located within one or more antenna arrays or antenna panels, which may support MIMO operations or transmit or receive beamforming. For example, one or more base station antennas or antenna arrays may be co-located at an antenna assembly, such as an antenna tower. In some examples, antennas or antenna arrays associated with a network entity **105** may be located at diverse geographic locations. A network entity **105** may include an antenna array with a set of rows and columns of antenna ports that the network entity **105** may use to support beamforming of communications with a UE **115**. Likewise, a UE **115** may include one or more antenna arrays that may support various MIMO or beamforming operations. Additionally, or alternatively, an antenna panel may support RF beamforming for a signal transmitted via an antenna port.

[0082] The network entities **105** or the UEs **115** may use MIMO communications to exploit multipath signal propagation and increase spectral efficiency by transmitting or receiving multiple signals via different spatial layers. Such techniques may be referred to as spatial multiplexing. The multiple signals may, for example, be transmitted by the transmitting device via different antennas or different combinations of antennas. Likewise, the multiple signals may be received by the receiving device via different antennas or different combinations of antennas. Each of the multiple signals may be referred to as a separate spatial stream and may carry information associated with the same data stream (e.g., the same codeword) or different data streams (e.g., different codewords). Different spatial layers may be associated with different antenna ports used for channel measurement and reporting. MIMO techniques include single-user MIMO (SU-MIMO), for which multiple spatial layers are transmitted to the same receiving device, and multiple-user MIMO (MU-MIMO), for which multiple spatial layers are transmitted to multiple devices.

[0083] Beamforming, which may also be referred to as spatial filtering, directional transmission, or directional

reception, is a signal processing technique that may be used at a transmitting device or a receiving device (e.g., a network entity **105**, a UE **115**) to shape or steer an antenna beam (e.g., a transmit beam, a receive beam) along a spatial path between the transmitting device and the receiving device. Beamforming may be achieved by combining the signals communicated via antenna elements of an antenna array such that some signals propagating along particular orientations with respect to an antenna array experience constructive interference while others experience destructive interference. The adjustment of signals communicated via the antenna elements may include a transmitting device or a receiving device applying amplitude offsets, phase offsets, or both to signals carried via the antenna elements associated with the device. The adjustments associated with each of the antenna elements may be defined by a beamforming weight set associated with a particular orientation (e.g., with respect to the antenna array of the transmitting device or receiving device, or with respect to some other orientation).

[0084] A network entity **105** or a UE **115** may use beam sweeping techniques as part of beamforming operations. For example, a network entity **105** (e.g., a base station **140**, an RU **170**) may use multiple antennas or antenna arrays (e.g., antenna panels) to conduct beamforming operations for directional communications with a UE **115**. Some signals (e.g., synchronization signals, reference signals, beam selection signals, or other control signals) may be transmitted by a network entity **105** multiple times along different directions. For example, the network entity **105** may transmit a signal according to different beamforming weight sets associated with different directions of transmission. Transmissions along different beam directions may be used to identify (e.g., by a transmitting device, such as a network entity **105**, or by a receiving device, such as a UE **115**) a beam direction for later transmission or reception by the network entity **105**.

[0085] Some signals, such as data signals associated with a particular receiving device, may be transmitted by a transmitting device (e.g., a network entity **105** or a UE **115**) along a single beam direction (e.g., a direction associated with the receiving device, such as another network entity **105** or UE **115**). In some examples, the beam direction associated with transmissions along a single beam direction may be determined based on a signal that was transmitted along one or more beam directions. For example, a UE **115** may receive one or more of the signals transmitted by the network entity **105** along different directions and may report to the network entity **105** an indication of the signal that the UE **115** received with a highest signal quality or an otherwise acceptable signal quality.

[0086] In some examples, transmissions by a device (e.g., by a network entity **105** or a UE **115**) may be performed using multiple beam directions, and the device may use a combination of digital precoding or beamforming to generate a combined beam for transmission (e.g., from a network entity **105** to a UE **115**). The UE **115** may report feedback that indicates precoding weights for one or more beam directions, and the feedback may correspond to a configured set of beams across a system bandwidth or one or more sub-bands. The network entity **105** may transmit a reference signal (e.g., a cell-specific reference signal (CRS), a channel state information reference signal (CSI-RS)), which may be precoded or unprecoded. The UE **115** may provide feedback for beam selection, which may be a precoding matrix indicator (PMI) or codebook-based feedback (e.g., a multi-

panel type codebook, a linear combination type codebook, a port selection type codebook). Although these techniques are described with reference to signals transmitted along one or more directions by a network entity **105** (e.g., a base station **140**, an RU **170**), a UE **115** may employ similar techniques for transmitting signals multiple times along different directions (e.g., for identifying a beam direction for subsequent transmission or reception by the UE **115**) or for transmitting a signal along a single direction (e.g., for transmitting data to a receiving device).

[0087] A receiving device (e.g., a UE **115**) may perform reception operations in accordance with multiple receive configurations (e.g., directional listening) when receiving various signals from a transmitting device (e.g., a network entity **105**), such as synchronization signals, reference signals, beam selection signals, or other control signals. For example, a receiving device may perform reception in accordance with multiple receive directions by receiving via different antenna subarrays, by processing received signals according to different antenna subarrays, by receiving according to different receive beamforming weight sets (e.g., different directional listening weight sets) applied to signals received at multiple antenna elements of an antenna array, or by processing received signals according to different receive beamforming weight sets applied to signals received at multiple antenna elements of an antenna array, any of which may be referred to as “listening” according to different receive configurations or receive directions. In some examples, a receiving device may use a single receive configuration to receive along a single beam direction (e.g., when receiving a data signal). The single receive configuration may be aligned along a beam direction determined based on listening according to different receive configurations (e.g., a beam direction determined to have a highest signal strength, highest signal-to-noise ratio (SNR), or otherwise acceptable signal quality based on listening according to multiple beam directions).

[0088] The wireless communications system **100** may be a packet-based network that operates according to a layered protocol stack. In the user plane, communications at the bearer or PDCP layer may be IP-based. An RLC layer may perform packet segmentation and reassembly to communicate via logical channels. A MAC layer may perform priority handling and multiplexing of logical channels into transport channels. The MAC layer also may implement error detection techniques, error correction techniques, or both to support retransmissions to improve link efficiency. In the control plane, an RRC layer may provide establishment, configuration, and maintenance of an RRC connection between a UE **115** and a network entity **105** or a core network **130** supporting radio bearers for user plane data. A PHY layer may map transport channels to physical channels.

[0089] The UEs **115** and the network entities **105** may support retransmissions of data to increase the likelihood that data is received successfully. Hybrid automatic repeat request (HARQ) feedback is one technique for increasing the likelihood that data is received correctly via a communication link (e.g., the communication link(s) **125**, a D2D communication link **135**). HARQ may include a combination of error detection (e.g., using a cyclic redundancy check (CRC)), forward error correction (FEC), and retransmission (e.g., automatic repeat request (ARQ)). HARQ may improve throughput at the MAC layer in relatively poor radio conditions (e.g., low signal-to-noise conditions). In some

examples, a device may support same-slot HARQ feedback, in which case the device may provide HARQ feedback in a specific slot for data received via a previous symbol in the slot. In some other examples, the device may provide HARQ feedback in a subsequent slot, or according to some other time interval.

[0090] Some wireless communication systems utilize reference signaling to characterize a channel. For example, a network entity **105** may transmit a set of reference signals. As used herein, a “set” may indicate a group of one or more items or elements. A UE **115** may receive the reference signaling and determine a channel estimate based on the reference signaling. The UE **115** may utilize the channel estimate to determine an equalizer for subsequent communications.

[0091] One challenge in wireless communications is to reduce the power consumption of the UE **115**. Equalizer coefficient calculation may consume a relatively large amount of processing resources at a UE receiver. In some approaches, the equalizer (e.g., LMMSE equalizer) coefficients are calculated per each frequency domain or time domain resource in a slot. Accordingly, a relatively large quantity of calculations may be carried out for coefficient calculation. Moreover, a calculation for a single frequency domain or time domain resource may be relatively costly regarding the quantity of multiplications because the calculation may involve matrix inversion and matrix multiplication. Further, some UEs **115** may support a relatively higher quantity of layers (e.g., spatial layers or MIMO layers). For example, instead of four layers, eight layers may be utilized. Accordingly, UEs **115** may have a greater quantity of receive antennas (e.g., eight antennas instead of four antennas) to accommodate more layers. Because the quantity of receive antennas may be utilized to determine the channel dimensions (e.g., H dimensions), the equalizer coefficient calculation complexity may increase with increasing channel dimensions. Even without an increasing the quantity of layers, power consumption for equalizer coefficients calculation is significant.

[0092] In some examples, decimation may be applied in the calculation of equalizer coefficients to reduce calculation complexity or power consumption. For instance, direct coefficient calculation may not be applied per resource (e.g., frequency domain or time domain resource). In some approaches, a UE **115** may perform calculations for frequency domain or time domain resources, but with one or more decimation factors. For example, a 2D decimation factor may be utilized for frequency and time dimensions. In some cases, a 2D decimation factor (e.g., frequency domain and time domain decimation factors) may significantly reduce the quantity of calculations. Accordingly, calculation complexity or power consumption may be significantly reduced. In some approaches, for the frequency domain or time domain resources that were decimated, the UE **115** may calculate equalizer coefficients with reduced complexity utilizing a linear interpolation between frequency domain or time domain resources.

[0093] Some examples of the techniques described herein may be utilized to reduce the complexity or power consumption of equalizer coefficient calculation. Some approaches may preserve performance (e.g., avoid or reduce performance loss) while reducing complexity or power

consumption. In some aspects, reference signaling (e.g., DMRS) overhead may be reduced or throughput may be increased.

[0094] FIG. 2 shows an example of a wireless communications system 200 that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure. The wireless communications system 200 may implement aspects of or may be implemented by aspects of the wireless communications system 100. For example, the wireless communications system 200 includes a UE 115-a, which may be an example of a UE 115 described with respect to FIG. 1. The wireless communications system 200 also includes a network entity 105-a, which may be an example of a network entity 105 as described with respect to FIG. 1.

[0095] The UE 115-a may communicate with the network entity 105-a using a communication link 125-a, which may be an example of a communication link 125 described with respect to FIG. 1. The communication link 125-a may include a bi-directional link that enables both uplink and downlink network communications. For example, the UE 115-a may transmit one or more uplink transmissions 205, such as uplink control signals or uplink data signals, to the network entity 105-a using the communication link 125-a, or the network entity 105-a may transmit one or more downlink transmissions 210, such as downlink control signals or downlink data signals, to the UE 115-a using the communication link 125-a.

[0096] The network entity 105-a may output, or the UE 115-a may receive, a set of reference signals 245 via a channel. Examples of the set of reference signals may include DMRSs, channel state information reference signals (CSI-RSs), tracking reference signals (TRSs), or other reference signals. In some examples, the set of reference signals 245 may be communicated (e.g., transmitted or received) via one or more channel resources (e.g., time resource, frequency resources, spatial resources, time slots, subcarriers, resource elements (REs), among other examples). In some aspects, the set of reference signals 245 may be communicated via an established 2D grid (e.g., a time domain and frequency domain grid). The set of reference signals 245 may be signals with one or more established properties (e.g., symbol(s), code(s), pattern(s), amplitude(s), frequency(ies), or timing(s)). For instance, the UE 115-a may store information indicating one or more of the established properties of one or more reference signals of the set of reference signals 245.

[0097] The set of reference signals may enable the UE 115-a to generate a set of channel estimates. For instance, the variation of a received reference signal with respect to one or more of the established properties of the reference signal may indicate the effect of the channel on the reference signal. A channel estimate may be information indicating one or more characteristics of the channel. In some examples, the channel estimate may indicate path loss, fading, phase, Doppler, signal-to-noise ratio (SNR), a quality metric, or another characteristic(s) of the channel.

[0098] For instance, the UE 115-a may utilize the set of reference signals 245 to generate (e.g., calculate or compute) channel estimates, where each channel estimate of the set of channel estimates may respectively correspond to a reference signal of the set of reference signals 245.

[0099] In some aspects of the techniques described herein, the UE 115-a may decimate the set of reference signals 245.

Decimating the set of reference signals 245 may include refraining from utilizing (e.g., may include removing, dropping, or deleting, among other examples) one or more of the set of reference signals 245 (for equalizer coefficient calculation, for instance).

[0100] A degree of the decimation may be based on the set of channel estimates. In some approaches, the degree of the decimation may be expressed as a decimation ratio in one or more dimensions. For example, a frequency-domain decimation ratio of 4 may indicate that one in four reference signals (or one reference signal over a set of four REs, for instance) in the frequency domain is utilized or that three out of four reference signals (or three REs in a set of four REs) in the frequency domain are not utilized for an equalizer coefficient calculation.

[0101] In some approaches, the UE 115-a may determine (e.g., estimate) the degree of decimation (e.g., one or more decimation ratios or factors) based on the set of reference signals 245. For example, the degree of decimation (e.g., one or more decimation ratios) may be controlled based on the channel between the UE 115-a and the network entity 105-a, which may be indicated by one or more channel estimates of the set of reference signals 245. In the frequency domain, the degree of decimation may be determined in an inverse relationship with channel dispersion. For instance, relatively less decimation may be performed with increasing channel dispersion. In the time domain, the degree of decimation may be determined in an inverse relationship with velocity (or phase). For instance, relatively less decimation may be performed with increasing velocity.

[0102] In some examples, determining the degree of decimation may include generating a measure of correlation between at least two channel estimates of the set of channel estimates. The UE 115-a may determine (e.g., calculate, compute) the measure of correlation. For instance, the measure of correlation may include a first measure in the time domain, a second measure in the frequency domain, or a combination thereof. In some approaches, the measure of correlation (e.g., $R(\tau)$) may be assessed in the time domain or the frequency domain in accordance with Equation (1).

$$R(\tau) = \frac{\sum_i \frac{1}{N_L N_R} \left(\sum_{i,j} H(i, j, t) H(i, j, t + \tau)^* \right)}{\sum_i \frac{1}{N_L N_R} \left(\sum_{i,j} \|H(i, j, t)\|^2 \right)} \quad (1)$$

In Equation (1), N_L denotes a quantity of layers, N_R denotes a quantity of antennas (e.g., receive antennas), and $H(i, j, t)$ denotes a channel estimate, where i is an antenna number or index (e.g., receive antenna index), j is a layer number or index, t is a symbol index or resource index, and (i, j) corresponds to a channel component in the matrix H . For a time domain correlation, the first measure in the time domain may be denoted $R_T(\tau)$, where t may denote a symbol index (e.g., OFDM symbol index) and τ may denote a time offset for measuring correlation. For a frequency domain correlation, the second measure in the frequency domain may be denoted $R_F(\tau)$, where t may denote a resource index (e.g., RE index) and τ may denote a frequency offset for measuring correlation.

[0103] The UE 115-a may determine the degree of decimation based on the measure of correlation. For instance, the UE 115-a may determine a degree of decimation in the time

domain based on the first measure $R_T(\tau)$ of correlation, may determine a degree of decimation in the frequency domain based on the second measure $R_F(\tau)$ of correlation, or a combination thereof.

[0104] In some aspects, the UE **115-a** may select the degree of decimation based on at least one threshold applied to the measure of correlation (e.g., applied to the first measure or the second measure). For example, the UE **115-a** may use a thresholding approach, where one or more correlations may be compared to one or more thresholds to determine the degree of decimation (e.g., decimation ratio). In some approaches, the at least one threshold includes a first set of (e.g., one or more) thresholds for a time domain and a second set of (e.g., one or more) thresholds for a frequency domain.

[0105] In some examples, the UE **115-a** may determine a decimation ratio in the time domain for $R_T(\tau=6)$ symbols) in accordance with Equation (2).

$$\text{Decimation ratio}_{TD} = \begin{cases} 4, & R_T(6) \geq \text{thr}_{TD,1} \\ 3, & \text{thr}_{TD,1} \geq R_T(6) \geq \text{thr}_{TD,2} \\ 2, & \text{thr}_{TD,2} \geq R_T(6) \geq \text{thr}_{TD,3} \\ 1, & \text{thr}_{TD,3} \geq R_T(6) \end{cases} \quad (2)$$

In Equation (2), Decimation ratio_{TD} denotes a decimation ratio in the time domain, $\text{thr}_{TD,1}$ denotes a first threshold for the time domain, $\text{thr}_{TD,2}$ denotes a second threshold for the time domain, and $\text{thr}_{TD,3}$ denotes a third threshold for the time domain, where $\text{thr}_{TD,1}$, $\text{thr}_{TD,2}$, and $\text{thr}_{TD,3}$, may be included in the first set of thresholds. While $\tau=6$ symbols, three thresholds, and four decimation ratios are given in the example of Equation (2), a different value of τ or fewer, more, or different decimation ratios, or quantities of thresholds may be utilized in other examples.

[0106] In some examples, the UE **115-a** may determine a decimation ratio in the frequency domain for $R_F(\tau=12)$ REs) in accordance with Equation (3).

$$\text{Decimation ratio}_{FD} = \begin{cases} 4, & R_F(12) \geq \text{thr}_{FD,1} \\ 2, & \text{thr}_{FD,1} \geq R_F(12) \geq \text{thr}_{FD,2} \\ 1, & \text{thr}_{FD,2} \geq R_F(12) \end{cases} \quad (3)$$

In Equation (3), Decimation ratio_{FD} denotes a decimation ratio in the frequency domain, $\text{thr}_{FD,1}$ denotes a first threshold for the frequency domain and $\text{thr}_{FD,2}$ denotes a second threshold for the frequency domain, where $\text{thr}_{FD,1}$ and $\text{thr}_{FD,2}$ may be included in the second set of thresholds. While $\tau=12$ REs, two thresholds, and three decimation ratios are given in the example of Equation (3), a different value of τ or fewer, more, or different decimation ratios, or quantities of thresholds may be utilized in other examples.

[0107] In some examples, the at least one threshold may include multiple sets of thresholds corresponding to respective modulation and coding schemes (MCSs). For instance, $\text{thr}_{TD,1}$, $\text{thr}_{TD,2}$, or $\text{thr}_{TD,3}$ may be determined (e.g., determined empirically, heuristically, or adaptively) per MCS. Table (1) provides examples of $\text{thr}_{TD,1}$, $\text{thr}_{TD,2}$, and $\text{thr}_{TD,3}$ for $R_T(\tau=6)$ symbols) for decimation ratio determination in the time domain.

TABLE 1

MCS	$\text{thr}_{TD,1}$	$\text{thr}_{TD,2}$	$\text{thr}_{TD,3}$
0	0.82	0.81	0.8
3	0.95	0.9	0.88
6	0.95	0.9	0.88
9	0.95	0.9	0.88
12	0.99	0.985	0.98
15	0.99	0.985	0.98
18	0.99	0.985	0.98
20	0.99	0.985	0.98
21	0.9998	0.9995	0.9993
22	0.9998	0.9995	0.9993
23	0.9998	0.9995	0.9993
24	0.9998	0.9995	0.9993
25	0.9998	0.9995	0.9993
26	0.9998	0.9995	0.9993
27	0.9998	0.9995	0.9993

In the example of Table (1), the thresholds may be determined per MCS by selecting values in terms of increased throughput.

[0108] In some examples, $\text{thr}_{FD,1}$ or $\text{thr}_{FD,2}$ may be determined (e.g., determined empirically, heuristically, or adaptively) per MCS. Table (2) provides examples of $\text{thr}_{FD,1}$ or $\text{thr}_{FD,2}$ for $R_F(\tau=12)$ REs) for decimation ratio determination in the frequency domain.

TABLE 2

MCS	$\text{thr}_{FD,1}$	$\text{thr}_{FD,2}$
0	0.81	0.78
3	0.81	0.78
6	0.81	0.78
9	0.81	0.78
12	0.91	0.88
15	0.91	0.88
18	0.91	0.88
20	0.91	0.88
21	0.91	0.88
22	0.91	0.88
23	0.91	0.88
24	0.98	0.88
25	0.98	0.88
26	0.98	0.89
27	0.98	0.93

In the example of Table (2), the thresholds may be determined per MCS by selecting values in terms of increased throughput.

[0109] The UE **115-a** may utilize the determined degree of decimation to decimate the set of reference signal **245**. In some approaches, the UE **115-a** may determine one or more sets of coefficients based on the decimated reference signals. For instance, the UE **115-a** may calculate a set of coefficients corresponding to each of the reference signals remaining after decimation or may refrain from calculating a coefficient(s) for each of the reference signals removed due to the decimation. The coefficients may be coefficients for an equalizer. For example, the UE **115-a** may calculate coefficients for an LMMSE equalizer for each of the remaining reference signals after decimation.

[0110] The UE **115-a** may interpolate the sets of coefficients. For instance, the UE **115-a** may perform an interpolation (e.g., linear interpolation or other interpolation) to determine one or more sets of coefficients for one or more resources between reference signals (e.g., between resources of the remaining reference signals after decimation). For

example, the UE 115-a may determine (e.g., calculate) a first set of coefficients corresponding to a first reference signal received via a first resource of the channel and may determine a second set of coefficients corresponding to a second reference signal received via a second resource of the channel. The first set of coefficients and the second set of coefficients may be calculated using a direct calculation (e.g., matrix inversion and matrix multiplication). The UE 115-a may interpolate the first set of coefficients and the second set of coefficients for a third resource of the channel between the first resource and the second resource to generate a third set of coefficients of the equalizer. A third reference signal corresponding to the third resource may have been decimated in accordance with the decimation. By interpolating the first set of coefficients and the second set of coefficients, the UE 115-a may determine a third set of coefficients using interpolation instead of directly calculating the third set of coefficients, which may avoid one or more complex operations (e.g., matrix inversion or matrix multiplication in the direct calculation).

[0111] The network entity 105-a may output, or the UE 115-a may receive, a signal 240 via the channel. For example, the signal 240 may be a data signal or a control signal from the network entity 105-a. The signal 240 may be equalized based on the decimation of the set of reference signals 245. For instance, the UE 115-a may generate an equalized signal from the signal 240 using an equalizer that is based on a decimation of the set of reference signals 245. For example, the UE 115-a may utilize the equalizer with interpolated coefficients to equalize the signal 240. Equalization may include reducing amplitude, magnitude, or response variation over a spectrum. For instance, the equalizer may attenuate or amplify one or more portions of the signal over a spectrum to reduce variations introduced by the channel or to flatten a spectral response. In some approaches, the equalized signal may be demodulated or decoded by the UE 115-a.

[0112] In some aspects, the UE 115-a may transmit, or the network entity 105-a may obtain, an indication 235 of the decimation. The indication 235 may be an explicit or implicit indication of the decimation. In some approaches, the indication 235 may be a parameter, a value, an index, a number, a signal timing, a signal arrangement, or other information indicating the decimation. In some examples, the indication 235 may indicate the degree of decimation (e.g., one or more decimation ratios) for the time domain or the frequency domain. In some examples, the indication 235 may indicate one or more resources (e.g., resource index(es)) of reference signals that are not removed in the decimation, or may indicate one or more resources (e.g., resource index(es)) of reference signals that are removed in the decimation.

[0113] In some approaches, the network entity 105-a may output, or the UE 115-a may receive, a request for the indication 235 of the decimation. For example, the network entity 105-a may transmit a parameter, value, or code indicating a request for the indication 235 of the decimation. In some examples, the network entity 105-a may request the indication 235 of equalizer decimation factors (e.g., a degree of decimation, a time domain decimation ratio, a frequency domain decimation ratio, or a combination thereof). For instance, the network entity 105-a may request that the UE 115-a indicate the estimated decimation factor(s) for use in the equalizer at the UE 115-a. In some examples, the

network entity 105-a (e.g., a gNB) may send the request (e.g., message) in the downlink over a physical downlink control channel (PDCCH) or a physical downlink shared channel (PDSCH) aperiodically (e.g., at aperiodic rate), based on a determination to refresh a report of the decimation.

[0114] The UE 115-a may transmit, or the network entity 105-a may receive, the indication 235 of the decimation in response to the request. For instance, the UE 115-a may reply with the indication 235, where the indication 235 may indicate equalizer decimation factors. In some aspects, the UE 115-a may reply with the current or latest estimated decimation factors for the time domain and the frequency domain. In some examples, the indication 235 (e.g., message) may be communicated via a physical uplink control channel (PUCCH) or a physical uplink shared channel (PUSCH).

[0115] In some aspects, the network entity 105-a determine, based on the indication 235 of the decimation, a quantity (e.g., density) for one or more subsequent sets of reference signals. For example, the network entity 105-a (e.g., gNB) may determine a reference signal (e.g., DMRS, CSI-RS, or TRS, among other examples) quantity in the time domain or frequency domain based on the indication 235 (e.g., the reported equalizer decimation factors). In some approaches, the network entity 105-a may translate (based on a policy, rule, or procedure, for instance) the reported frequency domain decimation ratio to a DMRS pattern, where the density per port of the DMRS pattern may be determined based on the frequency domain decimation ratio. Additionally, or alternatively, the network entity 105-a may translate (based on a policy, rule, or procedure, for instance), the reported time domain decimation ratio to a quantity of DMRSs.

[0116] The network entity 105-a may output, or the UE 115-a may receive, a second quantity of a second set of reference signals, where the second quantity is different from a first quantity of the set of reference signals 245. The second quantity may be based on the determined quantity or the indication 235 of the decimation. In some approaches, if the indication 235 indicates that the UE 115-a is decimating fewer reference signals (e.g., is decimating fewer than previously or is not decimating), the network entity 105-a may increase the quantity (e.g., density) of one or more subsequent sets of reference signals. If the indication 235 indicates that the UE 115-a is decimating more reference signals than the network entity 105-a has sent, the network entity 105-a may reduce the quantity (e.g., density) of one or more subsequent sets of reference signals. For example, the network entity 105-a may reduce the quantity of reference signals towards the quantity (or to match the quantity of reference signals being used) indicated by the indication 235 of the decimation. For instance, if the indication 235 of the decimation indicates that the UE 115-a is not utilizing $\frac{3}{4}$ (e.g., a decimation ratio of 4) of the reference signals for equalizer coefficient calculation, the network entity 105-a may reduce the quantity of reference signals to $\frac{1}{2}$ or $\frac{1}{4}$ of the previously output reference signal quantity. In some cases, the network entity 105-a may allocate one or more of the resources (previously used for reference signals) to output data, which may increase throughput.

[0117] In some examples, the network entity 105-a may provide downlink reference signaling (e.g., DMRS) with a diluted quantity of reference signals in the time domain or

the frequency domain (e.g., with reduced reference signal density). The network entity **105-a** may configure a subsequent downlink transmission to use the diluted reference signaling (e.g., DMRSs with diluted reference signaling in the frequency domain, time domain, or a combination thereof). The updated reference signal (e.g., DMRS) configuration may be encoded into a PDCCH, which may enable the UE **115-a** to decode the slot structure of the reference signaling.

[0118] In some approaches, the network entity **105-a** may output, or the UE **115-a** may receive, a request for an indication of a capability of the UE **115-a** to decimate the set of reference signals **245**. The request may be a request to indicate a capability of the UE **115-a** to perform decimation or a capability of the UE **115-a** to report the indication **235** of the decimation (e.g., equalizer decimation factors capability). The network entity **105-a** (e.g., gNB) may request the UE **115-a** to indicate whether the UE **115-a** has a capability to estimate the decimation factors for an equalizer or to transmit the indication **235** of the decimation. In some examples, the request for the indication of the capability may be communicated at a MAC layer (e.g., via a MAC control element (CE)). In some approaches, the request for the indication may be communicated during or after a cell attachment procedure (e.g., at or near the beginning of the communication between the UE **115-a** and the network entity **105-a**).

[0119] In some approaches, the UE **115-a** may transmit, or the network entity **105-a** may obtain, the indication of the capability of the UE **115-a** to decimate the set of reference signals **245**. For example, the indication of the capability of the UE **115-a** (e.g., the indication of the equalizer decimation factors capability) may be communicated in response to the request for the indication of the capability (of whether the UE **115-a** has a capability to estimate a degree of decimation or one or more decimation factors). In some examples, the UE **115-a** may transmit the indication of the capability without a request from the network entity **105-a**. In some aspects, the indication of the capability may be signaled as a bit indicating that the indication **235** of the decimation may also be sent at the MAC layer (e.g., at the MAC-CE level). In some approaches, the indication of the capability may be communicated during or after a cell attachment procedure (e.g., at or near the beginning of the communication between the UE **115-a** and the network entity **105-a**). In some approaches, one or more of the procedures described herein may be performed if the UE **115-a** indicates the capability to estimate the decimation or to report the indication **235** of the decimation.

[0120] Some examples of the techniques described herein may provide adaptive decimation based on one or more channel characteristics. In some scenarios, no performance loss may occur when applying the adaptive decimation ratios relative to no decimation. The UE **115-a** may signal the indication **235** of the decimation (e.g., decimation estimation) of one or more frequency domain or time domain decimation ratios. Based on the indication **235**, the network entity **105-a** may determine whether reference signaling (e.g., DMRS) may be diluted in the time domain, the frequency domain, or a combination thereof.

[0121] In some approaches, the network entity **105-a** may translate (based on a policy, rule, or procedure, for instance) the reported frequency domain decimation ratio to a reference signaling pattern (e.g., DMRS) with a frequency

domain density per port that is determined in accordance with the reported frequency domain decimation ratio. For example, when the frequency domain decimation factor is relatively high, the DMRS frequency domain density per port may be reduced or diluted.

[0122] In some approaches, the network entity **105-a** may translate (based on a policy, rule, or procedure, for instance) the reported time domain decimation ratio to a quantity of reference signals (e.g., DMRSs) in accordance with the reporting time domain decimation ratio. For example, when the time domain decimation factor is relatively high, the quantity of DMRSs may be reduced.

[0123] Some examples of the techniques described herein may reduce computation complexity or power consumption at the UE **115-a** by decimating the equalizer coefficients calculation. In some cases, the decimation may be performed without a performance loss or with a relatively small performance loss. Some of the techniques may enable reduced reference signaling (e.g., DMRS) overhead or increased throughput.

[0124] Some examples of the techniques may be utilized in difficult scenarios. For instance, the throughput performance of some of the techniques described herein may be comparable to throughput performance without decimation in scenarios with a dispersive channel and medium velocity (e.g., a tapped delay line-B (TDL-B) scenario with a delay spread of 250 nanoseconds (ns), a velocity=30 kilometers per hour (km/h), and 4 layers over a 4x4 channel). For example, decimation may be adapted as described herein to provide small or no throughput performance loss (e.g., to avoid performance loss) relative to a fixed decimation approach. In some scenarios, the attained decimation ratios in the time domain or frequency domain may be significant (and may depend on SNR). For example, a time domain decimation ratio that is less than or equal to 4 and a frequency domain decimation ratio less than or equal to 4 may provide decimation up to a factor of 16 with a minimum of 4. The decimation or reference signal adaptation may allow more resources to be allocated for data transmission, which may increase throughput performance.

[0125] FIG. 3 shows an example of a block diagram **300** that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure. In particular, FIG. 3 illustrates an example of a UE **115-b** that may be utilized in accordance with one or more of the techniques described herein. One or more of the elements of the UE **115-b** may be implemented in hardware or a combination of hardware and instructions or code (e.g., a processor with instructions). For example, the UE **115-b** may include a channel estimator **310**, a correlator **315**, a decimator **320**, a coefficient evaluator **335**, an interpolator **340**, or an equalizer **345**.

[0126] The UE **115-b** may receive a set of reference signals **305**. The set of reference signals **305** may be provided to the channel estimator **310**. The channel estimator **310** may generate a set of channel estimates corresponding to the set of reference signals as described with reference to FIG. 2. The set of channel estimates may be provided to the correlator **315**.

[0127] The correlator **315** may perform one or more correlation operations based on the set of channel estimates to determine one or more measures of correlation. For instance, the correlator **315** may correlate channel estimates over time or frequency as described with reference to FIG.

2 to produce a first measure of correlation in the time domain or a second measure of correlation in the frequency domain. The one or more measures of correlation may be provided to the decimator 320.

[0128] The decimator 320 may determine a degree of decimation based on the one or more measures of correlation. For example, the decimator 320 may compare a first measure $R_T(\tau)$ of correlation to a first set of thresholds 325 to determine a decimation ratio in the time domain. Additionally, or alternatively, the decimator 320 may compare a second measure $R_F(\tau)$ of correlation to a second set of thresholds 325 to determine a decimation ratio in the frequency domain. In some examples, the decimator 320 may produce an indication 330 of the decimation. The indication 330 of the decimation may be provided to the interpolator 340 or signaled to a network entity (e.g., network entity 105-a) in some approaches.

[0129] The decimator 320 may decimate the set of reference signals 305 based on the degree of decimation (e.g., the decimation ratio in the time domain, the decimation ratio in the frequency domain, or a combination thereof) to generate a set of decimated reference signals. For example, the decimator 320 may remove or drop a proportion of the set of reference signals 305 as described with reference to FIG. 2. The set of decimated reference signals may be provided to the coefficient evaluator 335.

[0130] The coefficient evaluator 335 may determine coefficients (e.g., one or more sets of coefficients) corresponding to the decimated reference signals. For example, the coefficient evaluator 335 may perform one or more operations (e.g., matrix inversions, matrix multiplication) to directly calculate the equalizer coefficients for each of the reference signals remaining after decimation as described with reference to FIG. 2. The coefficients may be provided to the interpolator 340.

[0131] The interpolator 340 may interpolate the coefficients to determine one or more interpolated coefficients (e.g., one or more sets of interpolated coefficients) based on the equalizer coefficients. For example, the interpolator 340 may interpolate one or more sets of coefficients for one or more resources corresponding to the removed reference signals as described with reference to FIG. 2. In some examples, the interpolation may be performed based on the indication 330 of the decimation. For instance, the indication 330 of the decimation may indicate a quantity of coefficients to interpolate between neighboring coefficients in accordance with one or more decimation ratios (e.g., a decimation ratio in the time domain, a decimation ratio in the frequency domain, or a combination thereof). The interpolator may provide the coefficients (e.g., interpolated coefficients and non-interpolated coefficients) to the equalizer 345.

[0132] The equalizer may utilize the coefficients to equalize a received signal 350. For example, the equalizer may equalize a signal received from a network entity as described with reference to FIG. 2 to generate an equalized signal.

[0133] FIG. 4 shows an example of a process flow 400 that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure. A wireless communication system may include a UE 115-c and a network entity 105-b. The UE 115-c may be an example of the UEs 115, the UE 115-a, or the UE 115-b, and the network entity 105-b may be an example of the network entities 105 or the network entity 105-a, as described herein.

[0134] In the following description of the process flow 400, the operations between the network entity 105-b and the UE 115-c may be transmitted in a different order than the example order shown, or the operations performed by the network entity 105-b and the UE 115-c may be performed in different orders or at different times. Some operations may be omitted from the process flow 400, and other operations may be added to the process flow 400. Further, although some operations or signaling may be shown to occur at different times for discussion purposes, these operations may actually occur at the same time or in overlapping time periods in some examples.

[0135] At 405, the network entity 105-b may output a capability request to the UE 115-c. For example, the UE 115-c may receive a capability request (e.g., an equalizer decimation factors capability query) for an indication of a capability of the UE 115-c to perform reference signal decimation or to report an indication of decimation as described with reference to FIG. 2 or FIG. 3.

[0136] At 410, the UE 115-c may transmit a capability indication to the network entity 105-b. For example, the UE 115-c may transmit a message (e.g., an equalizer decimation factors capability reply) indicating a capability of the UE 115-c to decimate reference signals or to report an indication of decimation as described with reference to FIG. 2 or FIG. 3.

[0137] At 415, the network entity 105-b may output a first set of reference signals. For example, the network entity 105-b may output a set of DMRSs to the UE 115-c as described with reference to FIG. 2 or FIG. 3.

[0138] At 420, the UE 115-c may perform channel estimation 420 based on the first set of reference signals. For example, the UE 115-c may calculate a set of channel estimates as described with reference to FIG. 2 or FIG. 3.

[0139] At 425, the UE 115-c may decimate the first set of reference signals. For instance, the UE 115-c may remove a portion of the first set of reference signals based on the set of channel estimates as described with reference to FIG. 2 or FIG. 3. In some examples, the UE 115-c may utilize the decimated reference signals to generate equalizer coefficients. The equalizer coefficients may be utilized to generate interpolated equalizer coefficients as described with reference to FIG. 2 or FIG. 3.

[0140] At 430, the network entity 105-b may output a signal to the UE 115-c. For example, the UE 115-c may receive a data signal or control signal from the network entity 105-b as described with reference to FIG. 2 or FIG. 3.

[0141] At 435, the UE 115-c may equalize the signal using an equalizer. For instance, the UE 115-c may equalize the signal based on the interpolated coefficients as described with reference to FIG. 2 or FIG. 3.

[0142] At 440, the network entity 105-b may output a decimation request. For example, the network entity 105-b may output a request for a decimation indication (e.g., a request for the decimation factor(s) of the equalizer at the UE 115-c) as described with reference to FIG. 2 or FIG. 3.

[0143] At 445, the UE 115-c may transmit an indication of the decimation. For example, the UE 115-c may transmit an indication of the decimation (e.g., a reply indicating the equalizer decimation ratio(s) or factor(s)) to the network entity 105-b as described with reference to FIG. 2 or FIG. 3.

[0144] At 450, the network entity 105-b may determine a quantity of reference signals based on the indication of the decimation. For instance, the network entity 105-b may

determine to increase or decrease a quantity of reference signals (e.g., DMRS time domain or frequency domain density based on the reported equalizer decimation factors) as described with reference to FIG. 2.

[0145] At 455, the network entity 105-b may output a second set of reference signals. For example, the network entity 105-b may output a second set of reference signals with more or fewer reference signals (e.g., diluted DMRS time domain or frequency domain density) as described with reference to FIG. 2.

[0146] FIG. 5 shows a block diagram 500 of a device 505 that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure. The device 505 may be an example of aspects of a UE 115 as described herein. The device 505 may include a receiver 510, a transmitter 515, and a communications manager 520. The device 505, or one or more components of the device 505 (e.g., the receiver 510, the transmitter 515, the communications manager 520), may include at least one processor, which may be coupled with at least one memory, to, individually or collectively, support or enable the described techniques. Each of these components may be in communication with one another (e.g., via one or more buses).

[0147] The receiver 510 may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to decimated reference signaling for equalization). Information may be passed on to other components of the device 505. The receiver 510 may utilize a single antenna or a set of multiple antennas.

[0148] The transmitter 515 may provide a means for transmitting signals generated by other components of the device 505. For example, the transmitter 515 may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to decimated reference signaling for equalization). In some examples, the transmitter 515 may be co-located with a receiver 510 in a transceiver module. The transmitter 515 may utilize a single antenna or a set of multiple antennas.

[0149] The communications manager 520, the receiver 510, the transmitter 515, or various combinations or components thereof may be examples of means for performing various aspects of decimated reference signaling for equalization as described herein. For example, the communications manager 520, the receiver 510, the transmitter 515, or various combinations or components thereof may be capable of performing one or more of the functions described herein.

[0150] In some examples, the communications manager 520, the receiver 510, the transmitter 515, or various combinations or components thereof may be implemented in hardware (e.g., in communications management circuitry). The hardware may include at least one of a processor, a digital signal processor (DSP), a central processing unit (CPU), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, a microcontroller, discrete gate or transistor logic, discrete hardware components, or any combination thereof configured as or otherwise supporting, individually or collectively, a means for performing the functions described in the present disclosure. In some

examples, at least one processor and at least one memory coupled with the at least one processor may be configured to perform one or more of the functions described herein (e.g., by one or more processors, individually or collectively, executing instructions stored in the at least one memory).

[0151] Additionally, or alternatively, the communications manager 520, the receiver 510, the transmitter 515, or various combinations or components thereof may be implemented in code (e.g., as communications management software or firmware) executed by at least one processor (e.g., referred to as a processor-executable code). If implemented in code executed by at least one processor, the functions of the communications manager 520, the receiver 510, the transmitter 515, or various combinations or components thereof may be performed by a general-purpose processor, a DSP, a CPU, an ASIC, an FPGA, a microcontroller, or any combination of these or other programmable logic devices (e.g., configured as or otherwise supporting, individually or collectively, a means for performing the functions described in the present disclosure).

[0152] In some examples, the communications manager 520 may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver 510, the transmitter 515, or both. For example, the communications manager 520 may receive information from the receiver 510, send information to the transmitter 515, or be integrated in combination with the receiver 510, the transmitter 515, or both to obtain information, output information, or perform various other operations as described herein.

[0153] For example, the communications manager 520 is capable of, configured to, or operable to support a means for receiving, from a network entity, a set of reference signals via a channel to generate a set of channel estimates, each channel estimate of the set of channel estimates respectively corresponding to a reference signal of the set of reference signals. The communications manager 520 is capable of, configured to, or operable to support a means for receiving a signal via the channel from the network entity. The communications manager 520 is capable of, configured to, or operable to support a means for generating an equalized signal from the signal using an equalizer that is based on a decimation of the set of reference signals, where a degree of the decimation is based on the set of channel estimates. The communications manager 520 is capable of, configured to, or operable to support a means for transmitting an indication of the decimation to the network entity.

[0154] By including or configuring the communications manager 520 in accordance with examples as described herein, the device 505 (e.g., at least one processor controlling or otherwise coupled with the receiver 510, the transmitter 515, the communications manager 520, or a combination thereof) may support techniques for reduced processing, reduced power consumption, or more efficient utilization of communication resources.

[0155] FIG. 6 shows a block diagram 600 of a device 605 that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure. The device 605 may be an example of aspects of a device 505 or a UE 115 as described herein. The device 605 may include a receiver 610, a transmitter 615, and a communications manager 620. The device 605, or one or more components of the device 605 (e.g., the receiver 610, the transmitter 615, the communications manager 620), may

include at least one processor, which may be coupled with at least one memory, to support the described techniques. Each of these components may be in communication with one another (e.g., via one or more buses).

[0156] The receiver 610 may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to decimated reference signaling for equalization). Information may be passed on to other components of the device 605. The receiver 610 may utilize a single antenna or a set of multiple antennas.

[0157] The transmitter 615 may provide a means for transmitting signals generated by other components of the device 605. For example, the transmitter 615 may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to decimated reference signaling for equalization). In some examples, the transmitter 615 may be co-located with a receiver 610 in a transceiver module. The transmitter 615 may utilize a single antenna or a set of multiple antennas.

[0158] The device 605, or various components thereof, may be an example of means for performing various aspects of decimated reference signaling for equalization as described herein. For example, the communications manager 620 may include a reference signal component 625, a signal component 630, an equalizer component 635, a decimation indication component 640, or any combination thereof. The communications manager 620 may be an example of aspects of a communications manager 520 as described herein. In some examples, the communications manager 620, or various components thereof, may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver 610, the transmitter 615, or both. For example, the communications manager 620 may receive information from the receiver 610, send information to the transmitter 615, or be integrated in combination with the receiver 610, the transmitter 615, or both to obtain information, output information, or perform various other operations as described herein.

[0159] The reference signal component 625 is capable of, configured to, or operable to support a means for receiving, from a network entity, a set of reference signals via a channel to generate a set of channel estimates, each channel estimate of the set of channel estimates respectively corresponding to a reference signal of the set of reference signals. The signal component 630 is capable of, configured to, or operable to support a means for receiving a signal via the channel from the network entity. The equalizer component 635 is capable of, configured to, or operable to support a means for generating an equalized signal from the signal using an equalizer that is based on a decimation of the set of reference signals, where a degree of the decimation is based on the set of channel estimates. The decimation indication component 640 is capable of, configured to, or operable to support a means for transmitting an indication of the decimation to the network entity.

[0160] FIG. 7 shows a block diagram 700 of a communications manager 720 that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure. The communications manager 720

may be an example of aspects of a communications manager 520, a communications manager 620, or both, as described herein. The communications manager 720, or various components thereof, may be an example of means for performing various aspects of decimated reference signaling for equalization as described herein. For example, the communications manager 720 may include a reference signal component 725, a signal component 730, an equalizer component 735, a decimation indication component 740, a capability component 745, a coefficient determination component 750, an interpolation component 755, a decimation determination component 760, or any combination thereof. Each of these components, or components or subcomponents thereof (e.g., one or more processors, one or more memories), may communicate, directly or indirectly, with one another (e.g., via one or more buses).

[0161] The reference signal component 725 is capable of, configured to, or operable to support a means for receiving, from a network entity, a set of reference signals via a channel to generate a set of channel estimates, each channel estimate of the set of channel estimates respectively corresponding to a reference signal of the set of reference signals. The signal component 730 is capable of, configured to, or operable to support a means for receiving a signal via the channel from the network entity. The equalizer component 735 is capable of, configured to, or operable to support a means for generating an equalized signal from the signal using an equalizer that is based on a decimation of the set of reference signals, where a degree of the decimation is based on the set of channel estimates. The decimation indication component 740 is capable of, configured to, or operable to support a means for transmitting an indication of the decimation to the network entity.

[0162] In some examples, the capability component 745 is capable of, configured to, or operable to support a means for receiving, from the network entity, a request for an indication of a capability of the UE to decimate the set of reference signals. In some examples, the capability component 745 is capable of, configured to, or operable to support a means for transmitting, to the network entity, the indication of the capability of the UE to decimate the set of reference signals.

[0163] In some examples, the decimation indication component 740 is capable of, configured to, or operable to support a means for receiving, from the network entity, a request for the indication of the decimation, where the indication of the decimation is transmitted in response to the request.

[0164] In some examples, the reference signal component 725 is capable of, configured to, or operable to support a means for receiving, from the network entity, a second quantity of a second set of reference signals, where the second quantity is different from a first quantity of the set of reference signals, where the second quantity is based on the indication of the decimation.

[0165] In some examples, the coefficient determination component 750 is capable of, configured to, or operable to support a means for determining a first set of coefficients corresponding to a first reference signal received via a first resource of the channel. In some examples, the coefficient determination component 750 is capable of, configured to, or operable to support a means for determining a second set of coefficients corresponding to a second reference signal received via a second resource of the channel. In some examples, the interpolation component 755 is capable of,

configured to, or operable to support a means for interpolating the first set of coefficients and the second set of coefficients for a third resource of the channel between the first resource and the second resource to generate a third set of coefficients of the equalizer, where a third reference signal corresponding to the third resource is decimated in accordance with the decimation.

[0166] In some examples, the decimation determination component 760 is capable of, configured to, or operable to support a means for determining the degree of decimation based on the set of reference signals.

[0167] In some examples, to support determining the degree of decimation, the decimation determination component 760 is capable of, configured to, or operable to support a means for generating a measure of correlation between at least two channel estimates of the set of channel estimates.

[0168] In some examples, the measure of correlation includes a first measure in a time domain, a second measure in a frequency domain, or a combination thereof.

[0169] In some examples, the decimation determination component 760 is capable of, configured to, or operable to support a means for selecting the degree of decimation based on at least one threshold applied to the measure of correlation.

[0170] In some examples, the at least one threshold includes a first set of thresholds for a time domain and a second set of thresholds for a frequency domain.

[0171] In some examples, the at least one threshold includes multiple sets of thresholds corresponding to respective MCSs.

[0172] FIG. 8 shows a diagram of a system 800 including a device 805 that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure. The device 805 may be an example of or include components of a device 505, a device 605, or a UE 115 as described herein. The device 805 may communicate (e.g., wirelessly) with one or more other devices (e.g., network entities 105, UEs 115, or a combination thereof). The device 805 may include components for bi-directional voice and data communications including components for transmitting and receiving communications, such as a communications manager 820, an input/output (I/O) controller, such as an I/O controller 810, a transceiver 815, one or more antennas 825, at least one memory 830, code 835, and at least one processor 840. These components may be in electronic communication or otherwise coupled (e.g., operatively, communicatively, functionally, electronically, electrically) via one or more buses (e.g., a bus 845).

[0173] The I/O controller 810 may manage input and output signals for the device 805. The I/O controller 810 may also manage peripherals not integrated into the device 805. In some cases, the I/O controller 810 may represent a physical connection or port to an external peripheral. In some cases, the I/O controller 810 may utilize an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®, LINUX®, or another known operating system. Additionally, or alternatively, the I/O controller 810 may represent or interact with a modem, a keyboard, a mouse, a touchscreen, or a similar device. In some cases, the I/O controller 810 may be implemented as part of one or more processors, such as the at least one processor 840. In some cases, a user may interact with the device 805 via the I/O controller 810 or via hardware components controlled by the I/O controller 810.

[0174] In some cases, the device 805 may include a single antenna. However, in some other cases, the device 805 may have more than one antenna, which may be capable of concurrently transmitting or receiving multiple wireless transmissions. The transceiver 815 may communicate bi-directionally via the one or more antennas 825 using wired or wireless links as described herein. For example, the transceiver 815 may represent a wireless transceiver and may communicate bi-directionally with another wireless transceiver. The transceiver 815 may also include a modem to modulate the packets, to provide the modulated packets to one or more antennas 825 for transmission, and to demodulate packets received from the one or more antennas 825. The transceiver 815, or the transceiver 815 and one or more antennas 825, may be an example of a transmitter 515, a transmitter 615, a receiver 510, a receiver 610, or any combination thereof or component thereof, as described herein.

[0175] The at least one memory 830 may include random access memory (RAM) and read-only memory (ROM). The at least one memory 830 may store computer-readable, computer-executable, or processor-executable code, such as the code 835. The code 835 may include instructions that, when executed by the at least one processor 840, cause the device 805 to perform various functions described herein. The code 835 may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some cases, the code 835 may not be directly executable by the at least one processor 840 but may cause a computer (e.g., when compiled and executed) to perform functions described herein. In some cases, the at least one memory 830 may include, among other things, a basic I/O system (BIOS) which may control basic hardware or software operation such as the interaction with peripheral components or devices.

[0176] The at least one processor 840 may include one or more intelligent hardware devices (e.g., one or more general-purpose processors, one or more DSPs, one or more central processing units (CPUs), one or more graphics processing units (GPUs), one or more neural processing units (NPU) (also referred to as neural network processors or deep learning processors (DLP)), one or more micro-controllers, one or more ASICs, one or more FPGAs, one or more programmable logic devices, discrete gate or transistor logic, one or more discrete hardware components, or any combination thereof). In some cases, the at least one processor 840 may be configured to operate a memory array using a memory controller. In some other cases, a memory controller may be integrated into the at least one processor 840. The at least one processor 840 may be configured to execute computer-readable instructions stored in a memory (e.g., the at least one memory 830) to cause the device 805 to perform various functions (e.g., functions or tasks supporting decimated reference signaling for equalization). For example, the device 805 or a component of the device 805 may include at least one processor 840 and at least one memory 830 coupled with or to the at least one processor 840, the at least one processor 840 and the at least one memory 830 configured to perform various functions described herein. In some examples, the at least one processor 840 may include multiple processors and the at least one memory 830 may include multiple memories. One or more of the multiple processors may be coupled with one or more of the multiple memories, which may, individually or

collectively, be configured to perform various functions described herein. In some examples, the at least one processor **840** may be a component of a processing system, which may refer to a system (such as a series) of machines, circuitry (including, for example, one or both of processor circuitry (which may include the at least one processor **840**) and memory circuitry (which may include the at least one memory **830**)), or components, that receives or obtains inputs and processes the inputs to produce, generate, or obtain a set of outputs. The processing system may be configured to perform one or more of the functions described herein. For example, the at least one processor **840** or a processing system including the at least one processor **840** may be configured to, configurable to, or operable to cause the device **805** to perform one or more of the functions described herein. Further, as described herein, being “configured to,” being “configurable to,” and being “operable to” may be used interchangeably and may be associated with a capability, when executing code **835** (e.g., processor-executable code) stored in the at least one memory **830** or otherwise, to perform one or more of the functions described herein.

[0177] For example, the communications manager **820** is capable of, configured to, or operable to support a means for receiving, from a network entity, a set of reference signals via a channel to generate a set of channel estimates, each channel estimate of the set of channel estimates respectively corresponding to a reference signal of the set of reference signals. The communications manager **820** is capable of, configured to, or operable to support a means for receiving a signal via the channel from the network entity. The communications manager **820** is capable of, configured to, or operable to support a means for generating an equalized signal from the signal using an equalizer that is based on a decimation of the set of reference signals, where a degree of the decimation is based on the set of channel estimates. The communications manager **820** is capable of, configured to, or operable to support a means for transmitting an indication of the decimation to the network entity.

[0178] By including or configuring the communications manager **820** in accordance with examples as described herein, the device **805** may support techniques for improved communication reliability, reduced latency, improved user experience related to reduced processing, reduced power consumption, more efficient utilization of communication resources, improved coordination between devices, longer battery life, improved utilization of processing capability.

[0179] In some examples, the communications manager **820** may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with the transceiver **815**, the one or more antennas **825**, or any combination thereof. Although the communications manager **820** is illustrated as a separate component, in some examples, one or more functions described with reference to the communications manager **820** may be supported by or performed by the at least one processor **840**, the at least one memory **830**, the code **835**, or any combination thereof. For example, the code **835** may include instructions executable by the at least one processor **840** to cause the device **805** to perform various aspects of decimated reference signaling for equalization as described herein, or the at least one processor **840** and the at least one memory **830** may be otherwise configured to, individually or collectively, perform or support such operations.

[0180] FIG. 9 shows a block diagram **900** of a device **905** that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure. The device **905** may be an example of aspects of a network entity **105** as described herein. The device **905** may include a receiver **910**, a transmitter **915**, and a communications manager **920**. The device **905**, or one or more components of the device **905** (e.g., the receiver **910**, the transmitter **915**, the communications manager **920**), may include at least one processor, which may be coupled with at least one memory, to, individually or collectively, support or enable the described techniques. Each of these components may be in communication with one another (e.g., via one or more buses).

[0181] The receiver **910** may provide a means for obtaining (e.g., receiving, determining, identifying) information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). Information may be passed on to other components of the device **905**. In some examples, the receiver **910** may support obtaining information by receiving signals via one or more antennas. Additionally, or alternatively, the receiver **910** may support obtaining information by receiving signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof.

[0182] The transmitter **915** may provide a means for outputting (e.g., transmitting, providing, conveying, sending) information generated by other components of the device **905**. For example, the transmitter **915** may output information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). In some examples, the transmitter **915** may support outputting information by transmitting signals via one or more antennas. Additionally, or alternatively, the transmitter **915** may support outputting information by transmitting signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof. In some examples, the transmitter **915** and the receiver **910** may be co-located in a transceiver, which may include or be coupled with a modem.

[0183] The communications manager **920**, the receiver **910**, the transmitter **915**, or various combinations or components thereof may be examples of means for performing various aspects of decimated reference signaling for equalization as described herein. For example, the communications manager **920**, the receiver **910**, the transmitter **915**, or various combinations or components thereof may be capable of performing one or more of the functions described herein.

[0184] In some examples, the communications manager **920**, the receiver **910**, the transmitter **915**, or various combinations or components thereof may be implemented in hardware (e.g., in communications management circuitry). The hardware may include at least one of a processor, a DSP, a CPU, an ASIC, an FPGA or other programmable logic device, a microcontroller, discrete gate or transistor logic, discrete hardware components, or any combination thereof configured as or otherwise supporting, individually or collectively, a means for performing the functions described in

the present disclosure. In some examples, at least one processor and at least one memory coupled with the at least one processor may be configured to perform one or more of the functions described herein (e.g., by one or more processors, individually or collectively, executing instructions stored in the at least one memory).

[0185] Additionally, or alternatively, the communications manager 920, the receiver 910, the transmitter 915, or various combinations or components thereof may be implemented in code (e.g., as communications management software or firmware) executed by at least one processor (e.g., referred to as a processor-executable code). If implemented in code executed by at least one processor, the functions of the communications manager 920, the receiver 910, the transmitter 915, or various combinations or components thereof may be performed by a general-purpose processor, a DSP, a CPU, an ASIC, an FPGA, a microcontroller, or any combination of these or other programmable logic devices (e.g., configured as or otherwise supporting, individually or collectively, a means for performing the functions described in the present disclosure).

[0186] In some examples, the communications manager 920 may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver 910, the transmitter 915, or both. For example, the communications manager 920 may receive information from the receiver 910, send information to the transmitter 915, or be integrated in combination with the receiver 910, the transmitter 915, or both to obtain information, output information, or perform various other operations as described herein.

[0187] For example, the communications manager 920 is capable of, configured to, or operable to support a means for outputting, to a UE, a set of reference signals via a channel. The communications manager 920 is capable of, configured to, or operable to support a means for outputting, to the UE via the channel, a signal for equalization based on a decimation of the set of reference signals. The communications manager 920 is capable of, configured to, or operable to support a means for obtaining, from the UE, an indication of the decimation of the set of reference signals.

[0188] By including or configuring the communications manager 920 in accordance with examples as described herein, the device 905 (e.g., at least one processor controlling or otherwise coupled with the receiver 910, the transmitter 915, the communications manager 920, or a combination thereof) may support techniques for reduced processing, reduced power consumption, or more efficient utilization of communication resources.

[0189] FIG. 10 shows a block diagram 1000 of a device 1005 that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure. The device 1005 may be an example of aspects of a device 905 or a network entity 105 as described herein. The device 1005 may include a receiver 1010, a transmitter 1015, and a communications manager 1020. The device 1005, or one or more components of the device 1005 (e.g., the receiver 1010, the transmitter 1015, the communications manager 1020), may include at least one processor, which may be coupled with at least one memory, to support the described techniques. Each of these components may be in communication with one another (e.g., via one or more buses).

[0190] The receiver 1010 may provide a means for obtaining (e.g., receiving, determining, identifying) information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). Information may be passed on to other components of the device 1005. In some examples, the receiver 1010 may support obtaining information by receiving signals via one or more antennas. Additionally, or alternatively, the receiver 1010 may support obtaining information by receiving signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof.

[0191] The transmitter 1015 may provide a means for outputting (e.g., transmitting, providing, conveying, sending) information generated by other components of the device 1005. For example, the transmitter 1015 may output information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). In some examples, the transmitter 1015 may support outputting information by transmitting signals via one or more antennas. Additionally, or alternatively, the transmitter 1015 may support outputting information by transmitting signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof. In some examples, the transmitter 1015 and the receiver 1010 may be co-located in a transceiver, which may include or be coupled with a modem.

[0192] The device 1005, or various components thereof, may be an example of means for performing various aspects of decimated reference signaling for equalization as described herein. For example, the communications manager 1020 may include a reference signal manager 1025, a signal manager 1030, a decimation manager 1035, or any combination thereof. The communications manager 1020 may be an example of aspects of a communications manager 920 as described herein. In some examples, the communications manager 1020, or various components thereof, may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver 1010, the transmitter 1015, or both. For example, the communications manager 1020 may receive information from the receiver 1010, send information to the transmitter 1015, or be integrated in combination with the receiver 1010, the transmitter 1015, or both to obtain information, output information, or perform various other operations as described herein.

[0193] The reference signal manager 1025 is capable of, configured to, or operable to support a means for outputting, to a UE, a set of reference signals via a channel. The signal manager 1030 is capable of, configured to, or operable to support a means for outputting, to the UE via the channel, a signal for equalization based on a decimation of the set of reference signals. The decimation manager 1035 is capable of, configured to, or operable to support a means for obtaining, from the UE, an indication of the decimation of the set of reference signals.

[0194] FIG. 11 shows a block diagram 1100 of a communications manager 1120 that supports decimated reference signaling for equalization in accordance with one or more

aspects of the present disclosure. The communications manager 1120 may be an example of aspects of a communications manager 920, a communications manager 1020, or both, as described herein. The communications manager 1120, or various components thereof, may be an example of means for performing various aspects of decimated reference signaling for equalization as described herein. For example, the communications manager 1120 may include a reference signal manager 1125, a signal manager 1130, a decimation manager 1135, a capability manager 1140, a decimation indication manager 1145, or any combination thereof. Each of these components, or components or sub-components thereof (e.g., one or more processors, one or more memories), may communicate, directly or indirectly, with one another (e.g., via one or more buses). The communications may include communications within a protocol layer of a protocol stack, communications associated with a logical channel of a protocol stack (e.g., between protocol layers of a protocol stack, within a device, component, or virtualized component associated with a network entity 105, between devices, components, or virtualized components associated with a network entity 105), or any combination thereof.

[0195] The reference signal manager 1125 is capable of, configured to, or operable to support a means for outputting, to a UE, a set of reference signals via a channel. The signal manager 1130 is capable of, configured to, or operable to support a means for outputting, to the UE via the channel, a signal for equalization based on a decimation of the set of reference signals. The decimation manager 1135 is capable of, configured to, or operable to support a means for obtaining, from the UE, an indication of the decimation of the set of reference signals.

[0196] In some examples, the capability manager 1140 is capable of, configured to, or operable to support a means for outputting, to the UE, a request for an indication of a capability of the UE to decimate the set of reference signals. In some examples, the capability manager 1140 is capable of, configured to, or operable to support a means for obtaining, from the UE, the indication of the capability of the UE to decimate the set of reference signals.

[0197] In some examples, the decimation indication manager 1145 is capable of, configured to, or operable to support a means for outputting, to the UE, a request for the indication of the decimation, where the indication of the decimation is obtained in response to the request.

[0198] In some examples, the reference signal manager 1125 is capable of, configured to, or operable to support a means for determining, based on the indication of the decimation, a quantity for a second set of reference signals.

[0199] In some examples, the reference signal manager 1125 is capable of, configured to, or operable to support a means for outputting, to the UE, a second quantity of the second set of reference signals, where the second quantity is different from a first quantity of the set of reference signals, where the second quantity is based on the quantity.

[0200] FIG. 12 shows a diagram of a system 1200 including a device 1205 that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure. The device 1205 may be an example of or include components of a device 905, a device 1005, or a network entity 105 as described herein. The device 1205 may communicate with other network devices or network equipment such as one or more of the network

entities 105, UEs 115, or any combination thereof. The communications may include communications over one or more wired interfaces, over one or more wireless interfaces, or any combination thereof. The device 1205 may include components that support outputting and obtaining communications, such as a communications manager 1220, a transceiver 1210, one or more antennas 1215, at least one memory 1225, code 1230, and at least one processor 1235. These components may be in electronic communication or otherwise coupled (e.g., operatively, communicatively, functionally, electronically, electrically) via one or more buses (e.g., a bus 1240).

[0201] The transceiver 1210 may support bi-directional communications via wired links, wireless links, or both as described herein. In some examples, the transceiver 1210 may include a wired transceiver and may communicate bi-directionally with another wired transceiver. Additionally, or alternatively, in some examples, the transceiver 1210 may include a wireless transceiver and may communicate bi-directionally with another wireless transceiver. In some examples, the device 1205 may include one or more antennas 1215, which may be capable of transmitting or receiving wireless transmissions (e.g., concurrently). The transceiver 1210 may also include a modem to modulate signals, to provide the modulated signals for transmission (e.g., by one or more antennas 1215, by a wired transmitter), to receive modulated signals (e.g., from one or more antennas 1215, from a wired receiver), and to demodulate signals. In some implementations, the transceiver 1210 may include one or more interfaces, such as one or more interfaces coupled with the one or more antennas 1215 that are configured to support various receiving or obtaining operations, or one or more interfaces coupled with the one or more antennas 1215 that are configured to support various transmitting or outputting operations, or a combination thereof. In some implementations, the transceiver 1210 may include or be configured for coupling with one or more processors or one or more memory components that are operable to perform or support operations based on received or obtained information or signals, or to generate information or other signals for transmission or other outputting, or any combination thereof. In some implementations, the transceiver 1210, or the transceiver 1210 and the one or more antennas 1215, or the transceiver 1210 and the one or more antennas 1215 and one or more processors or one or more memory components (e.g., the at least one processor 1235, the at least one memory 1225, or both), may be included in a chip or chip assembly that is installed in the device 1205. In some examples, the transceiver 1210 may be operable to support communications via one or more communications links (e.g., communication link(s) 125, backhaul communication link(s) 120, a midhaul communication link 162, a fronthaul communication link 168).

[0202] The at least one memory 1225 may include RAM, ROM, or any combination thereof. The at least one memory 1225 may store computer-readable, computer-executable, or processor-executable code, such as the code 1230. The code 1230 may include instructions that, when executed by one or more of the at least one processor 1235, cause the device 1205 to perform various functions described herein. The code 1230 may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some cases, the code 1230 may not be directly executable by a processor of the at least one processor 1235

but may cause a computer (e.g., when compiled and executed) to perform functions described herein. In some cases, the at least one memory 1225 may include, among other things, a BIOS which may control basic hardware or software operation such as the interaction with peripheral components or devices. In some examples, the at least one processor 1235 may include multiple processors and the at least one memory 1225 may include multiple memories. One or more of the multiple processors may be coupled with one or more of the multiple memories which may, individually or collectively, be configured to perform various functions herein (for example, as part of a processing system).

[0203] The at least one processor 1235 may include one or more intelligent hardware devices (e.g., one or more general-purpose processors, one or more DSPs, one or more central processing units (CPUs), one or more graphics processing units (GPUs), one or more neural processing units (NPU) (also referred to as neural network processors or deep learning processors (DLPs)), one or more micro-controllers, one or more ASICs, one or more FPGAs, one or more programmable logic devices, discrete gate or transistor logic, one or more discrete hardware components, or any combination thereof). In some cases, the at least one processor 1235 may be configured to operate a memory array using a memory controller. In some other cases, a memory controller may be integrated into one or more of the at least one processor 1235. The at least one processor 1235 may be configured to execute computer-readable instructions stored in a memory (e.g., one or more of the at least one memory 1225) to cause the device 1205 to perform various functions (e.g., functions or tasks supporting decimated reference signaling for equalization). For example, the device 1205 or a component of the device 1205 may include at least one processor 1235 and at least one memory 1225 coupled with one or more of the at least one processor 1235, the at least one processor 1235 and the at least one memory 1225 configured to perform various functions described herein. The at least one processor 1235 may be an example of a cloud-computing platform (e.g., one or more physical nodes and supporting software such as operating systems, virtual machines, or container instances) that may host the functions (e.g., by executing code 1230) to perform the functions of the device 1205. The at least one processor 1235 may be any one or more suitable processors capable of executing scripts or instructions of one or more software programs stored in the device 1205 (such as within one or more of the at least one memory 1225). In some examples, the at least one processor 1235 may include multiple processors and the at least one memory 1225 may include multiple memories. One or more of the multiple processors may be coupled with one or more of the multiple memories, which may, individually or collectively, be configured to perform various functions herein. In some examples, the at least one processor 1235 may be a component of a processing system, which may refer to a system (such as a series) of machines, circuitry (including, for example, one or both of processor circuitry (which may include the at least one processor 1235) and memory circuitry (which may include the at least one memory 1225)), or components, that receives or obtains inputs and processes the inputs to produce, generate, or obtain a set of outputs. The processing system may be configured to perform one or more of the functions described herein. For example, the at least one processor 1235 or a processing system including the at least one

processor 1235 may be configured to, configurable to, or operable to cause the device 1205 to perform one or more of the functions described herein. Further, as described herein, being “configured to,” being “configurable to,” and being “operable to” may be used interchangeably and may be associated with a capability, when executing code stored in the at least one memory 1225 or otherwise, to perform one or more of the functions described herein.

[0204] In some examples, a bus 1240 may support communications of (e.g., within) a protocol layer of a protocol stack. In some examples, a bus 1240 may support communications associated with a logical channel of a protocol stack (e.g., between protocol layers of a protocol stack), which may include communications performed within a component of the device 1205, or between different components of the device 1205 that may be co-located or located in different locations (e.g., where the device 1205 may refer to a system in which one or more of the communications manager 1220, the transceiver 1210, the at least one memory 1225, the code 1230, and the at least one processor 1235 may be located in one of the different components or divided between different components).

[0205] In some examples, the communications manager 1220 may manage aspects of communications with a core network 130 (e.g., via one or more wired or wireless backhaul links). For example, the communications manager 1220 may manage the transfer of data communications for client devices, such as one or more UEs 115. In some examples, the communications manager 1220 may manage communications with one or more other network entities 105, and may include a controller or scheduler for controlling communications with UEs 115 (e.g., in cooperation with the one or more other network devices). In some examples, the communications manager 1220 may support an X2 interface within an LTE/LTE-A wireless communications network technology to provide communication between network entities 105.

[0206] For example, the communications manager 1220 is capable of, configured to, or operable to support a means for outputting, to a UE, a set of reference signals via a channel. The communications manager 1220 is capable of, configured to, or operable to support a means for outputting, to the UE via the channel, a signal for equalization based on a decimation of the set of reference signals. The communications manager 1220 is capable of, configured to, or operable to support a means for obtaining, from the UE, an indication of the decimation of the set of reference signals.

[0207] By including or configuring the communications manager 1220 in accordance with examples as described herein, the device 1205 may support techniques for improved communication reliability, reduced latency, reduced power consumption, more efficient utilization of communication resources, improved coordination between devices, longer battery life, or improved utilization of processing capability.

[0208] In some examples, the communications manager 1220 may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the transceiver 1210, the one or more antennas 1215 (e.g., where applicable), or any combination thereof. Although the communications manager 1220 is illustrated as a separate component, in some examples, one or more functions described with reference to the communications manager 1220 may be sup-

ported by or performed by the transceiver **1210**, one or more of the at least one processor **1235**, one or more of the at least one memory **1225**, the code **1230**, or any combination thereof (for example, by a processing system including at least a portion of the at least one processor **1235**, the at least one memory **1225**, the code **1230**, or any combination thereof). For example, the code **1230** may include instructions executable by one or more of the at least one processor **1235** to cause the device **1205** to perform various aspects of decimated reference signaling for equalization as described herein, or the at least one processor **1235** and the at least one memory **1225** may be otherwise configured to, individually or collectively, perform or support such operations.

[0209] FIG. **13** shows a flowchart illustrating a method **1300** that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure. The operations of the method **1300** may be implemented by a UE or its components as described herein. For example, the operations of the method **1300** may be performed by a UE **115** as described with reference to FIGS. **1** through **8**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.

[0210] At **1305**, the method may include receiving, from a network entity, a set of reference signals via a channel to generate a set of channel estimates, each channel estimate of the set of channel estimates respectively corresponding to a reference signal of the set of reference signals. The operations of **1305** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1305** may be performed by a reference signal component **725** as described with reference to FIG. **7**.

[0211] At **1310**, the method may include receiving a signal via the channel from the network entity. The operations of **1310** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1310** may be performed by a signal component **730** as described with reference to FIG. **7**.

[0212] At **1315**, the method may include generating an equalized signal from the signal using an equalizer that is based on a decimation of the set of reference signals, where a degree of the decimation is based on the set of channel estimates. The operations of **1315** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1315** may be performed by an equalizer component **735** as described with reference to FIG. **7**.

[0213] At **1320**, the method may include transmitting an indication of the decimation to the network entity. The operations of **1320** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1320** may be performed by a decimation indication component **740** as described with reference to FIG. **7**.

[0214] FIG. **14** shows a flowchart illustrating a method **1400** that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure. The operations of the method **1400** may be implemented by a UE or its components as described herein. For example, the operations of the method **1400** may be performed by a UE **115** as described with reference to FIGS. **1** through **8**. In some examples, a UE may execute a set of

instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.

[0215] At **1405**, the method may include receiving, from the network entity, a request for an indication of a capability of the UE to decimate the set of reference signals. The operations of **1405** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1405** may be performed by a capability component **745** as described with reference to FIG. **7**.

[0216] At **1410**, the method may include transmitting, to the network entity, the indication of the capability of the UE to decimate the set of reference signals. The operations of **1410** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1410** may be performed by a capability component **745** as described with reference to FIG. **7**.

[0217] At **1415**, the method may include receiving, from a network entity, a set of reference signals via a channel to generate a set of channel estimates, each channel estimate of the set of channel estimates respectively corresponding to a reference signal of the set of reference signals. The operations of **1415** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1415** may be performed by a reference signal component **725** as described with reference to FIG. **7**.

[0218] At **1420**, the method may include receiving a signal via the channel from the network entity. The operations of **1420** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1420** may be performed by a signal component **730** as described with reference to FIG. **7**.

[0219] At **1425**, the method may include generating an equalized signal from the signal using an equalizer that is based on a decimation of the set of reference signals, where a degree of the decimation is based on the set of channel estimates. The operations of **1425** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1425** may be performed by an equalizer component **735** as described with reference to FIG. **7**.

[0220] At **1430**, the method may include transmitting an indication of the decimation to the network entity. The operations of **1430** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1430** may be performed by a decimation indication component **740** as described with reference to FIG. **7**.

[0221] FIG. **15** shows a flowchart illustrating a method **1500** that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure. The operations of the method **1500** may be implemented by a network entity or its components as described herein. For example, the operations of the method **1500** may be performed by a network entity as described with reference to FIGS. **1** through **4** and **9** through **12**. In some examples, a network entity may execute a set of instructions to control the functional elements of the network entity to perform the described functions. Additionally, or alternatively, the network entity may perform aspects of the described functions using special-purpose hardware.

[0222] At **1505**, the method may include outputting, to a UE, a set of reference signals via a channel. The operations

of **1505** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1505** may be performed by a reference signal manager **1125** as described with reference to FIG. **11**.

[0223] At **1510**, the method may include outputting, to the UE via the channel, a signal for equalization based on a decimation of the set of reference signals. The operations of **1510** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1510** may be performed by a signal manager **1130** as described with reference to FIG. **11**.

[0224] At **1515**, the method may include obtaining, from the UE, an indication of the decimation of the set of reference signals. The operations of **1515** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1515** may be performed by a decimation manager **1135** as described with reference to FIG. **11**.

[0225] FIG. **16** shows a flowchart illustrating a method **1600** that supports decimated reference signaling for equalization in accordance with one or more aspects of the present disclosure. The operations of the method **1600** may be implemented by a network entity or its components as described herein. For example, the operations of the method **1600** may be performed by a network entity as described with reference to FIGS. **1** through **4** and **9** through **12**. In some examples, a network entity may execute a set of instructions to control the functional elements of the network entity to perform the described functions. Additionally, or alternatively, the network entity may perform aspects of the described functions using special-purpose hardware.

[0226] At **1605**, the method may include outputting, to the UE, a request for an indication of a capability of the UE to decimate the set of reference signals. The operations of **1605** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1605** may be performed by a capability manager **1140** as described with reference to FIG. **11**.

[0227] At **1610**, the method may include obtaining, from the UE, the indication of the capability of the UE to decimate the set of reference signals. The operations of **1610** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1610** may be performed by a capability manager **1140** as described with reference to FIG. **11**.

[0228] At **1615**, the method may include outputting, to a UE, a set of reference signals via a channel. The operations of **1615** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1615** may be performed by a reference signal manager **1125** as described with reference to FIG. **11**.

[0229] At **1620**, the method may include outputting, to the UE via the channel, a signal for equalization based on a decimation of the set of reference signals. The operations of **1620** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1620** may be performed by a signal manager **1130** as described with reference to FIG. **11**.

[0230] At **1625**, the method may include obtaining, from the UE, an indication of the decimation of the set of reference signals. The operations of **1625** may be performed in accordance with examples as disclosed herein. In some

examples, aspects of the operations of **1625** may be performed by a decimation manager **1135** as described with reference to FIG. **11**.

[0231] The following provides an overview of aspects of the present disclosure:

[0232] Aspect 1: A method for wireless communications at a UE, comprising: receiving, from a network entity, a set of reference signals via a channel to generate a set of channel estimates, each channel estimate of the set of channel estimates respectively corresponding to a reference signal of the set of reference signals; receiving a signal via the channel from the network entity; generating an equalized signal from the signal using an equalizer that is based at least in part on a decimation of the set of reference signals, wherein a degree of the decimation is based at least in part on the set of channel estimates; and transmitting an indication of the decimation to the network entity.

[0233] Aspect 2: The method of aspect 1, further comprising: receiving, from the network entity, a request for an indication of a capability of the UE to decimate the set of reference signals; and transmitting, to the network entity, the indication of the capability of the UE to decimate the set of reference signals.

[0234] Aspect 3: The method of any of aspects 1 through 2, further comprising: receiving, from the network entity, a request for the indication of the decimation, wherein the indication of the decimation is transmitted in response to the request.

[0235] Aspect 4: The method of any of aspects 1 through 3, further comprising: receiving, from the network entity, a second quantity of a second set of reference signals, wherein the second quantity is different from a first quantity of the set of reference signals, wherein the second quantity is based at least in part on the indication of the decimation.

[0236] Aspect 5: The method of any of aspects 1 through 4, further comprising: determining a first set of coefficients corresponding to a first reference signal received via a first resource of the channel; determining a second set of coefficients corresponding to a second reference signal received via a second resource of the channel; and interpolating the first set of coefficients and the second set of coefficients for a third resource of the channel between the first resource and the second resource to generate a third set of coefficients of the equalizer, wherein a third reference signal corresponding to the third resource is decimated in accordance with the decimation.

[0237] Aspect 6: The method of any of aspects 1 through 5, further comprising: determining the degree of decimation based at least in part on the set of reference signals.

[0238] Aspect 7: The method of aspect 6, wherein determining the degree of decimation comprises: generating a measure of correlation between at least two channel estimates of the set of channel estimates.

[0239] Aspect 8: The method of aspect 7, wherein the measure of correlation comprises a first measure in a time domain, a second measure in a frequency domain, or a combination thereof.

[0240] Aspect 9: The method of any of aspects 7 through 8, further comprising: selecting the degree of decimation based at least in part on at least one threshold applied to the measure of correlation.

[0241] Aspect 10: The method of aspect 9, wherein the at least one threshold comprises a first set of thresholds for a time domain and a second set of thresholds for a frequency domain.

[0242] Aspect 11: The method of any of aspects 9 through 10, wherein the at least one threshold comprises multiple sets of thresholds corresponding to respective MCSs.

[0243] Aspect 12: A method for wireless communications at a network entity, comprising: outputting, to a UE, a set of reference signals via a channel; outputting, to the UE via the channel, a signal for equalization based at least in part on a decimation of the set of reference signals; and obtaining, from the UE, an indication of the decimation of the set of reference signals.

[0244] Aspect 13: The method of aspect 12, further comprising: outputting, to the UE, a request for an indication of a capability of the UE to decimate the set of reference signals; and obtaining, from the UE, the indication of the capability of the UE to decimate the set of reference signals.

[0245] Aspect 14: The method of any of aspects 12 through 13, further comprising: outputting, to the UE, a request for the indication of the decimation, wherein the indication of the decimation is obtained in response to the request.

[0246] Aspect 15: The method of any of aspects 12 through 14, further comprising: determining, based at least in part on the indication of the decimation, a quantity for a second set of reference signals.

[0247] Aspect 16: The method of aspect 15, further comprising: outputting, to the UE, a second quantity of the second set of reference signals, wherein the second quantity is different from a first quantity of the set of reference signals, wherein the second quantity is based at least in part on the quantity.

[0248] Aspect 17: A UE comprising one or more memories storing processor-executable code, and one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the UE to perform a method of any of aspects 1 through 11.

[0249] Aspect 18: A UE comprising at least one means for performing a method of any of aspects 1 through 11.

[0250] Aspect 19: A non-transitory computer-readable medium storing code, the code comprising instructions executable by one or more processors to perform a method of any of aspects 1 through 11.

[0251] Aspect 20: A network entity comprising one or more memories storing processor-executable code, and one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the network entity to perform a method of any of aspects 12 through 16.

[0252] Aspect 21: A network entity comprising at least one means for performing a method of any of aspects 12 through 16.

[0253] Aspect 22: A non-transitory computer-readable medium storing code, the code comprising instructions executable by one or more processors to perform a method of any of aspects 12 through 16.

[0254] It should be noted that the methods described herein describe possible implementations. The operations and the steps may be rearranged or otherwise modified and other implementations are possible. Further, aspects from two or more of the methods may be combined.

[0255] Although aspects of an LTE, LTE-A, LTE-A Pro, or NR system may be described for purposes of example, and LTE, LTE-A, LTE-A Pro, or NR terminology may be used in much of the description, the techniques described herein are applicable beyond LTE, LTE-A, LTE-A Pro, or NR networks. For example, the described techniques may be applicable to various other wireless communications systems such as Ultra Mobile Broadband (UMB), Institute of Electrical and Electronics Engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, Flash-OFDM, as well as other systems and radio technologies not explicitly mentioned herein.

[0256] Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0257] The various illustrative blocks and components described in connection with the disclosure herein may be implemented or performed using a general-purpose processor, a DSP, an ASIC, a CPU, a graphics processing unit (GPU), a neural processing unit (NPU), an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor but, in the alternative, the processor may be any processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration). Any functions or operations described herein as being capable of being performed by a processor may be performed by multiple processors that, individually or collectively, are capable of performing the described functions or operations.

[0258] The functions described herein may be implemented using hardware, software executed by a processor, firmware, or any combination thereof. If implemented using software executed by a processor, the functions may be stored as or transmitted using one or more instructions or code of a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described herein may be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

[0259] Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one location to another. A non-transitory storage medium may be any available medium that may be accessed by a general-purpose or special-purpose computer. By way of example, and not limitation, non-transitory computer-readable media may include RAM, ROM, electrically erasable programmable ROM (EEPROM), flash memory, compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage

devices, or any other non-transitory medium that may be used to carry or store desired program code means in the form of instructions or data structures and that may be accessed by a general-purpose or special-purpose computer or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of computer-readable medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk, and Blu-ray disc. Disks may reproduce data magnetically, and discs may reproduce data optically using lasers. Combinations of the above are also included within the scope of computer-readable media. Any functions or operations described herein as being capable of being performed by a memory may be performed by multiple memories that, individually or collectively, are capable of performing the described functions or operations.

[0260] As used herein, including in the claims, “or” as used in a list of items (e.g., a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an example step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

[0261] As used herein, including in the claims, the article “a” before a noun is open-ended and understood to refer to “at least one” of those nouns or “one or more” of those nouns. Thus, the terms “a,” “at least one,” “one or more,” and “at least one of one or more” may be interchangeable. For example, if a claim recites “a component” that performs one or more functions, each of the individual functions may be performed by a single component or by any combination of multiple components. Thus, the term “a component” having characteristics or performing functions may refer to “at least one of one or more components” having a particular characteristic or performing a particular function. Subsequent reference to a component introduced with the article “a” using the terms “the” or “said” may refer to any or all of the one or more components. For example, a component introduced with the article “a” may be understood to mean “one or more components,” and referring to “the component” subsequently in the claims may be understood to be equivalent to referring to “at least one of the one or more components.” Similarly, subsequent reference to a component introduced as “one or more components” using the terms “the” or “said” may refer to any or all of the one or more components. For example, referring to “the one or more components” subsequently in the claims may be understood to be equivalent to referring to “at least one of the one or more components.”

[0262] The term “determine” or “determining” encompasses a variety of actions and, therefore, “determining” can

include calculating, computing, processing, deriving, investigating, looking up (such as via looking up in a table, a database, or another data structure), ascertaining, and the like. Also, “determining” can include receiving (e.g., receiving information), accessing (e.g., accessing data stored in memory), and the like. Also, “determining” can include resolving, obtaining, selecting, choosing, establishing, and other such similar actions.

[0263] In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label or other subsequent reference label.

[0264] The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “example” used herein means “serving as an example, instance, or illustration” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some figures, known structures and devices are shown in block diagram form in order to avoid obscuring the concepts of the described examples.

[0265] The description herein is provided to enable a person having ordinary skill in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to a person having ordinary skill in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

What is claimed is:

1. A user equipment (UE), comprising:

one or more memories storing processor-executable code; and

one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the UE to:

receive, from a network entity, a set of reference signals via a channel to generate a set of channel estimates, each channel estimate of the set of channel estimates respectively corresponding to a reference signal of the set of reference signals;

receive a signal via the channel from the network entity;

generate an equalized signal from the signal using an equalizer that is based at least in part on a decimation of the set of reference signals, wherein a degree of the decimation is based at least in part on the set of channel estimates; and

transmit an indication of the decimation to the network entity.

2. The UE of claim 1, wherein the one or more processors are individually or collectively further operable to execute the code to cause the UE to:

receive, from the network entity, a request for an indication of a capability of the UE to decimate the set of reference signals; and

transmit, to the network entity, the indication of the capability of the UE to decimate the set of reference signals.

3. The UE of claim 1, wherein the one or more processors are individually or collectively further operable to execute the code to cause the UE to:

receive, from the network entity, a request for the indication of the decimation, wherein the indication of the decimation is transmitted in response to the request.

4. The UE of claim 1, wherein the one or more processors are individually or collectively further operable to execute the code to cause the UE to:

receive, from the network entity, a second quantity of a second set of reference signals, wherein the second quantity is different from a first quantity of the set of reference signals, wherein the second quantity is based at least in part on the indication of the decimation.

5. The UE of claim 1, wherein the one or more processors are individually or collectively further operable to execute the code to cause the UE to:

determine a first set of coefficients corresponding to a first reference signal received via a first resource of the channel;

determine a second set of coefficients corresponding to a second reference signal received via a second resource of the channel; and

interpolate the first set of coefficients and the second set of coefficients for a third resource of the channel between the first resource and the second resource to generate a third set of coefficients of the equalizer, wherein a third reference signal corresponding to the third resource is decimated in accordance with the decimation.

6. The UE of claim 1, wherein the one or more processors are individually or collectively further operable to execute the code to cause the UE to:

determine the degree of decimation based at least in part on the set of reference signals.

7. The UE of claim 6, wherein, to determine the degree of decimation, the one or more processors are individually or collectively operable to execute the code to cause the UE to:

generate a measure of correlation between at least two channel estimates of the set of channel estimates.

8. The UE of claim 7, wherein the measure of correlation comprises a first measure in a time domain, a second measure in a frequency domain, or a combination thereof.

9. The UE of claim 7, wherein the one or more processors are individually or collectively further operable to execute the code to cause the UE to:

select the degree of decimation based at least in part on at least one threshold applied to the measure of correlation.

10. The UE of claim 9, wherein the at least one threshold comprises a first set of thresholds for a time domain and a second set of thresholds for a frequency domain.

11. The UE of claim 9, wherein:

the at least one threshold comprises multiple sets of thresholds corresponding to respective modulation and coding schemes.

12. A network entity, comprising:

one or more memories storing processor-executable code; and

one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the network entity to:

output, to a user equipment (UE), a set of reference signals via a channel;

output, to the UE via the channel, a signal for equalization based at least in part on a decimation of the set of reference signals; and

obtain, from the UE, an indication of the decimation of the set of reference signals.

13. The network entity of claim 12, wherein the one or more processors are individually or collectively further operable to execute the code to cause the network entity to:

output, to the UE, a request for an indication of a capability of the UE to decimate the set of reference signals; and

obtain, from the UE, the indication of the capability of the UE to decimate the set of reference signals.

14. The network entity of claim 12, wherein the one or more processors are individually or collectively further operable to execute the code to cause the network entity to:

output, to the UE, a request for the indication of the decimation, wherein the indication of the decimation is obtained in response to the request.

15. The network entity of claim 12, wherein the one or more processors are individually or collectively further operable to execute the code to cause the network entity to:

determine, based at least in part on the indication of the decimation, a quantity for a second set of reference signals.

16. The network entity of claim 15, wherein the one or more processors are individually or collectively further operable to execute the code to cause the network entity to:

output, to the UE, a second quantity of the second set of reference signals, wherein the second quantity is different from a first quantity of the set of reference signals, wherein the second quantity is based at least in part on the quantity.

17. A method for wireless communications at a user equipment (UE), comprising:

receiving, from a network entity, a set of reference signals via a channel to generate a set of channel estimates, each channel estimate of the set of channel estimates respectively corresponding to a reference signal of the set of reference signals;

receiving a signal via the channel from the network entity;

generating an equalized signal from the signal using an equalizer that is based at least in part on a decimation of the set of reference signals, wherein a degree of the decimation is based at least in part on the set of channel estimates; and

transmitting an indication of the decimation to the network entity.

18. The method of claim 17, further comprising:

receiving, from the network entity, a request for an indication of a capability of the UE to decimate the set of reference signals; and

transmitting, to the network entity, the indication of the capability of the UE to decimate the set of reference signals.

19. The method of claim **17**, further comprising:
receiving, from the network entity, a request for the
indication of the decimation, wherein the indication of
the decimation is transmitted in response to the request.

20. The method of claim **17**, further comprising:
receiving, from the network entity, a second quantity of a
second set of reference signals, wherein the second
quantity is different from a first quantity of the set of
reference signals, wherein the second quantity is based
at least in part on the indication of the decimation.

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